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LANL PJMIF Concept
Proprietary Report

S. Woodruff, A. Higginbottom, S. Ward

March 25, 2025

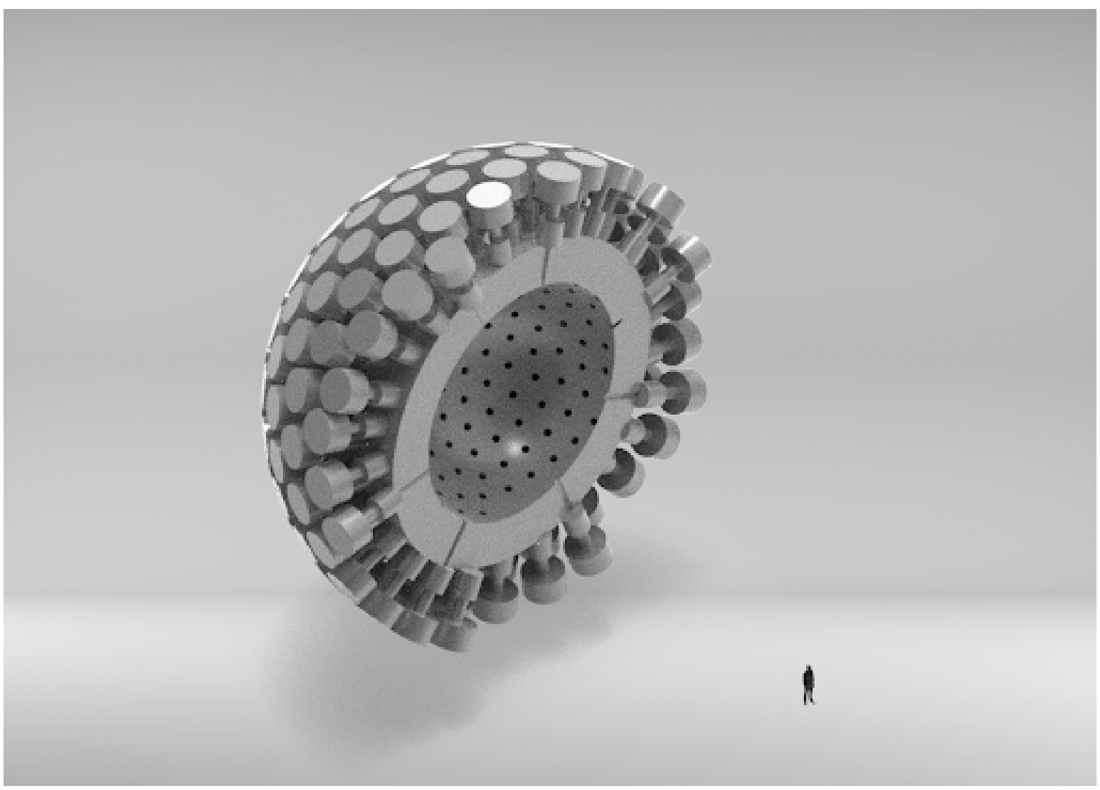

Figure 1: Plasma Jet Magneto-Inertial Fusion (PJMIF) Concept

Foreword

In the last several years, Woodruff Scientific has maintained an active interest in the costing analysis for fusion energy systems, collaborating with many in the field, and publishing a series of papers on this topic starting in 2011 with basic community consensus on the 'Path to Market' for fusion energy systems, outlining the constraints placed on any energy technology as it moves towards commercialization in terms of timeline and available capital, and market demand for smaller, more modular units [1]. We followed up this analysis with a specific embodiment of a compact modular fusion power core that could be developed privately [2], working in the context of an IAEA Coordinated Research Project on compact fusion neutron sources and developing a costing model based on the ARIES methodology. In 2017, we found support from ARPA-E to further develop our costing model in collaboration with Bechtel [3]. For that costing study, we considered 4 of the concepts that were supported by ARPA-E under the ALPHA program, and worked with PIs in each organization to develop basic radial builds of the 4 fusion power cores, based on prior art, and then performed cost analysis with the Balance of Plant and tritium systems analysis provided by Bechtel, but based predominantly on historical studies. In 2018 Woodruff organized a workshop for the IAEA on private fusion enterprises [4], and the commercialization path for fusion, with contributions from most major private fusion institutions and provided a summary of the cost modeling underway.

In 2019 ARPA-E revisited the 2017 costing analysis in the light of new costing paradigms developed by Ingersoll and Foss at Lucid Catalyst [5] [6], addressing every assumption in the Cost Accounting Structure (CAS), calculating an updated \$/kW for most CAS outside of the fusion power core (drawing in particular on the DOE NETL report [7]). The 2019 ARPA-E study revisited the analysis using the code developed for the four concepts, touching on most of the recommendations of the prior study and obtaining review by leading experts in the field. The costing was revisited in 2020 to extend to all ARPA-E supported fusion concepts, and the final version was started in 2023 to be released publicly. This study captures the most recent thinking and brings together all our understanding of best costing methodologies uncovered in the couse of the ARPA-E work.

Simon Woodruff

President

Woodruff Scientific LTD

Contents

1 Executive Summary

This report contains a Standard Costing Analysis, performed with the ARPA-E Fusion Economics Code (pyFECONS). It also contains appendices that augment the standard analysis. The principal findings are summarized here.

The economics of the fusion energy system proposed by Los Alamos look good: a 200MWe scale DT power plant comes in at $ 1.4Bn providing a LCOE at 10.5c/kWh. While this is already comparable to the currently projected competitive 5c/kWh range, we note that there are several factors that allow this cost to be brought down further, which will naturally fall out of a pre-conceptual design study which will impact the cost of pulsed power, buildings, structure, target delivery, and other components in the power core. Other factors that impact installation time could include use of modular construction that is manufactured centrally, thereby reducing indirect costs. It is easy to imagine this cost falling further.

There are several other factors that are important to consider and these are captured as follows:

- The pulse frequency of 1Hz has been assumed throughout. However, stepping on the gas gives an immediate reduction on LCOE. Going to 3Hz brings the LCOE down to 7.6c/kWh. Such pulse frequencies need to be examined in detail at the pre-conceptual design level for feasibility.
- The wall plug efficiency as been set at 10 % for both compression and target formation, which if increased to 25%, the LCOE falls from 10.5c/kWh to 9.2c/kWh.
- Learning curves are not assumed for the guns and power supplies - we have taken a $1/Watt installed cost but if we assume a 20% learning is possible for the power supplies, which are the dominant cost driver, reducing costs by x 2, then the LCOE falls form 10.5 to 10.0c/kWh.
- Considering the construction time, shortening from 2 years to 6 months installation reduces the LCOE from 10.5 to 10.2c/kWh.

In combination, the pulse frequency, wall plug efficiency, shorter installation time and reduction in pulsed power, we can bring the costs down from 105c/kWh to 6c/kWh. Note that this is slightly high relative to what is customarily thought competitive (aiming for less than 5c/kWh). There are paths towards the optimization of the design that we can perform as part of a 'cost driver analysis'.

- Deep cost driver analysis for the pulsed power systems, and chamber, to include scaling of chamber size to reach wall loading limts, and interdependence of major cost categories;
- Analysis of the implications of Reliability, Availability, Maintainability and Inspectability, and how these compare with other configurations;
- Implications for materials selections on cost - are there cost savings to be made by selecting certain materials?
- Examination of the indirect costs, which appear to be large.

Building a digital twin of the concept by building complete FEM models of the major system elements would be an important next step beyond the cost driver analysis. This fidelity of modeling would allow component selections to be made that reduce cost, and identify trade-offs.

2 Methodology

In this cost report, we draw on three major reports that summarize the cost categories for a power plant, based on the methodology developed first by the IAEA, then by the GENIV Economics Modeling Working Group (G4EMWG), and by the lead of the G4EMWG. Geoffrey Rothwell, in his book 'Economics of Nuclear Power'. There is purposeful use of existing well-documented and contemporary cost basis methodologies represented in these reports, as this documentation will assist anyone who wishes to perform similar analysis.

The cost categories depart from prior fusion costing studies by breaking the direct costs into categories 10 (pre-construction) and 20 (construction) and the indirect costs into categories 30-60 (formerly 91-98). There are some major similarities (in 20, for example, they are almost exactly the same, except for inclusion of a digital twin and contingency), and there are minor variations of the categories that make a lot more sense, for example decommissioning costs are included as a capitalized indirect cost rather than added somewhat obscurely to the LCOE, allowing deeper discussion of the contributing costs to decommissioning. In this section we describe the main texts that are foundational to the methodology and also those that provide information on cost basis. Fusion systems of course depart significantly from those of fission in the heat island, fuel cycle, handling, and replacement schedule of major cost components - those are spelled out in each section. We adopt the subcategory descriptions of most of the GENIV subcategories, but have edited those to more narrowly capture the fusion system costs.

Our philosophy has been to provide recent cost basis information for all cost categories, and state those in the text, providing a reference to the cost basis, and any other reference that will assist the user of the code. We have also tried to be judicious in our use of terminology: at the time of writing, the term 'reactor' is still viewed unfavorably and it's continued use may cause hurdles in the adoption of fusion energy systems, so we have replaced with 'heat island'. For old hands, and those in the nuclear sector, this change in terminology may seem to be superficial, since currently there is renewed interest in nuclear power as a solution for climate mitigation. We will evolve the terms as necessary, but for this edition, we try to be consistent in the use of terminology that conveys function without connotations.

Figure 2: Cost categories outlined below

2.1 Documents influencing the costing methodology

The GENIV Economics Modeling Working Group (G4EMWG) Guidelines [8] is a comprehensive guide providing detailed guidelines for economic modeling within the context of Generation IV nuclear energy systems. It systematically outlines the structure for cost estimation, covering various cost categories ranging from pre-construction costs to annualized financial costs. These include detailed breakdowns for capitalized costs (like direct costs, indirect service costs, owner's costs, and supplementary costs), as well as operational and maintenance costs, fuel costs, and financial costs. The document is meticulously organized, offering a granular view of each cost component, including land rights, permits, licensing, structures, equipment, and contingency costs. It serves as an essential resource for accurately assessing the economic feasibility and overall budgeting for advanced nuclear power plant projects. The guidelines emphasize the importance of detailed cost accounting for ensuring financial viability and strategic planning in the nuclear energy sector.

The book "The Economics of Future Nuclear Power", by Geofrey Rothwell [9], is a comprehensive analysis of the economic factors influencing the future of nuclear power. The document broadly delves into various aspects such as cost estimation, risk aversion, and capital cost determination for nuclear power projects. It discusses the intricacies of financial modeling and risk assessment, emphasizing the importance of accurately gauging uncertainties and potential costs associated with nuclear energy development. The text also explores the concept of risk diversification, applying financial market theories to the realm of nuclear power investment and development. The content is quite technical, focusing on economic and financial theories as they apply to the nuclear power industry, with detailed mathematical and statistical analyses to support its discussions.

The IAEA Tecdoc TRS396 [10] is a technical report by the International Atomic Energy Agency (IAEA), providing comprehensive guidelines and methodologies for economic evaluation in the nuclear power sector. The report delves into the economic analysis of nuclear power plants, offering detailed methods for calculating the Levelized Discounted Electricity Generation Costs (LDEGC). It incorporates a range of financial aspects including capital investment costs, operation and maintenance costs, and nuclear fuel cycle costs. The report also discusses the significance of various economic parameters such as discount rates, inflation, and escalation rates in the assessment of nuclear power projects. Additionally, it offers insights into sensitivity analysis, which is crucial for understanding the impact of varying economic conditions on project viability. The document is a valuable resource for professionals in the nuclear industry, providing a structured approach to evaluating the economic feasibility of nuclear power projects.

2.2 Documents providing cost bases

The UCSD Report titled "ARIES Cost Account Documentation" by L. M. Waganer, dated June 2013, from the University of California San Diego's Center for Energy Research [11], is a comprehensive documentation of the ARIES Cost Account. Its purpose is to document the historical economic basis for the ARIES Systems Code costing analyses and develop an updated economic model for future systems studies. It provides a thorough background on previous fusion studies' costing information, detailing conceptual designs and cost data from projects like Starfire, Generomak, and various Inertial Fusion Energy (IFE) designs. The report discusses the importance of historical cost escalation, explaining how past estimates are adjusted to current economic conditions using factors like the U.S. GDP Implicit Price Deflator. It also revisits the general cost account information, originally defined by DOE through the Pacific Northwest Laboratory, and covers various aspects such as spare parts, contingency, and Level of Safety Assurance (LSA) in cost estimation. The document offers a detailed analysis of direct and indirect capital costs, elaborating on each cost account with respect to fusion power plants and their respective costing algorithms, adjustments, and recommendations.

The presentation titled "Progress On Cost Modeling - Cost Accounts 20 and 21" by L. M. Waganer, dated 14-15 June 2007 [12] is centered on the ARIES Project and focuses on improving and updating cost modeling for fusion power plant designs. It discusses the need for revising the ARIES systems code cost modeling, originally based on models from the early 1990s, due to lack of extensive documentation and limited depth in published cost data. The document highlights various cost accounts, like Cryogenic Cooling, Radioactive Waste Treatment, and Reactor Plant Maintenance Equipment, and stresses the necessity for their detailed documentation. The presentation also touches upon general costing ground rules, safety assurance levels, and provides recommendations for defining and reporting cost accounts with reasonable depth. It includes specific details about costs related to land rights, structures, and site facilities for a fusion power plant, proposing updated cost estimates and methodologies for more accurate and comprehensive financial planning in the context of advanced nuclear fusion research.

The document "NETL Cost and Performance Baseline for Fossil Energy Plants Volume 1: Bituminous Coal and Natural Gas to Electricity" [7] provides an independent assessment of the cost and performance of selected fossil energy power systems. These systems include Integrated Gasification Combined Cycle (IGCC), Pulverized Coal (PC), and Natural Gas Combined Cycle (NGCC) plants. The assessment is conducted using a systematic, transparent technical, and economic approach. This report is the first in a four-volume series, each addressing different aspects of fossil fuel-based power generation. It plays a critical role in assessing and determining technology combinations for future power markets, offering insights for technology comparisons, and providing a framework for regulators and policymakers. In total, nineteen power plant configurations are analyzed, including various IGCC configurations with and without CO2 capture, PC power plant configurations both subcritical and supercritical with varying levels of CO2 capture, and NGCC power plant configurations with state-of-the-art combustion turbines, again with varying levels of CO2 capture.

2.3 Prior related costing studies

This concept was originally investigated in 2017 as part of the ARPA-E ALPHA costing study [3]. There we examined the capital costs but did not extend to the Levelized Cost of Electricity. A figure of the original concept is shown.

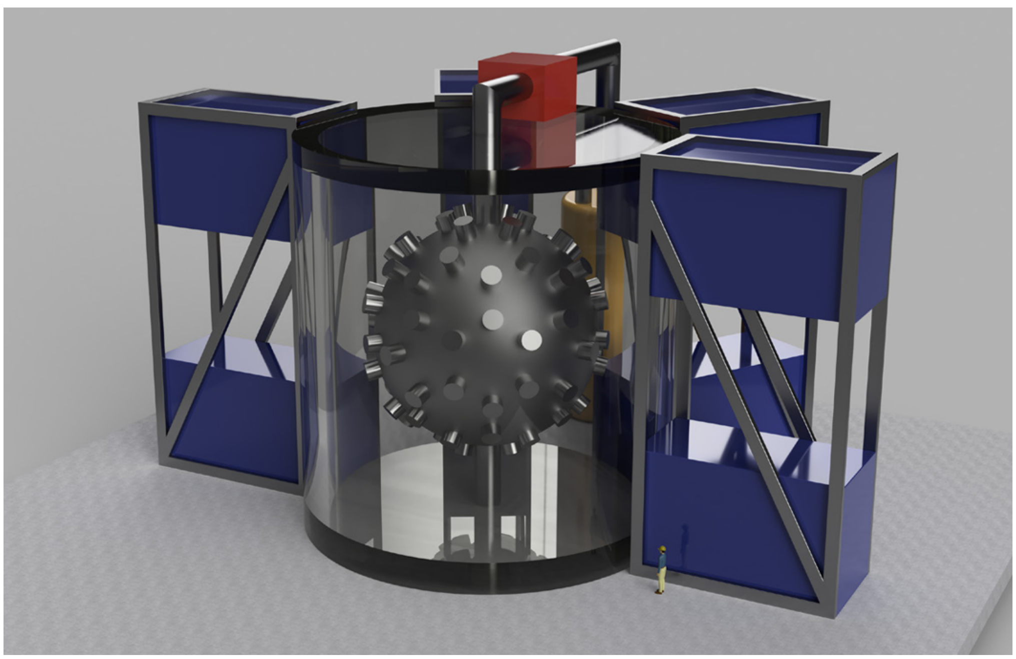

Figure 3: Original ARPA-E costing performed with Bechtel

We include here an appendix containing the first of two reports developed to justify the cost bases for the fusion power core with Los Alamos. This report contains a comprehensive bibliogrpahy.

3 Power balance

A representative power-flow diagram for a fusion-based electric power station is presented in Fig. 4. Power, P_{IN}, is input to the fusing plasma with efficiency, η_{IN}. For the D-T fuel cycle, the fusion power, P_F, is partitioned 20 % to 3.52-MeV alpha particles and 80 % to 14.06-MeV neutrons, is relatable to the input power by the gain, Q, as a figure of merit. In a Li-bearing blanket, the neutron power is multiplied by a factor, $M_n \simeq 1.1$. The useful thermal power is P_{TH}, which is converted by a turbine-generator (TG) set to a gross electric power, P_{ET}. A fraction of P_{ET} is recirculated to provide P_{IN}/η_{IN} as well as pumping power for the primary coolant and other housekeeping power. The total recirculating power fraction is ϵ, a figure

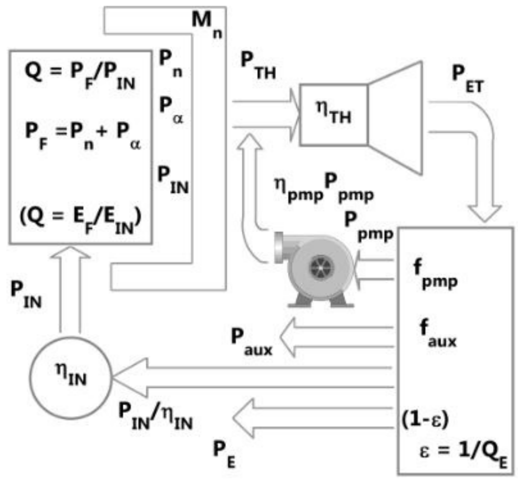

Figure 4: Fusion-based electric power station power-flow diagram.

of merit that monitors site power. The net electric power available for sale to the grid is $P_E = (1-\epsilon)P_{ET}$. Not shown on the diagram is the lost power, $P_{Loss} = (1-\eta_{TH})P_{TH} + (1-\eta_{IN})P_{IN}/\eta_{IN}$. An alternative definition, $Q_E = (1-\epsilon)/\epsilon$, is sometimes used. In the desirable limit of low ϵ, these expressions converge asymptotically. Note that the latter form is what was used recently by Wurzel and Hsu [13].

Given a stipulated target for the net electric-power output, P_E, the thermal-power output, P_{TH}, is determined for a value of the thermal conversion efficiency, η_{TH}, such that $P_E = (1-\epsilon)\eta_{TH}P_{TH}$, where $\epsilon = 1/Q_E$ is the recirculating power fraction and Q_E is the engineering gain. The gross electric-power output is $P_{ET} = \eta_{TH}P_{TH}$. A fraction f_{aux} of P_{ET} ($P_{aux} = f_{aux}P_{ET}$) is allocated for auxiliary functions (coolant and housekeeping); a fraction of the gross electric power is allocated to cryo systems, P_{cryo}. A fraction f_{pump} of P_{ET} ($P_{pump} = f_{pump}P_{ET}$) is allocated for primary-loop pumping power. It is assumed that $\eta_{pump}\,P_{pmp}$ is recoverable as useful thermal power in the primary coolant loop. These fractions of the gross electric power, including also the input power (P_{IN}) are commonly referred to as recirculating power. The engineering gain, Q_E, can be written as:

$$Q_E \quad = \quad \frac{1}{\epsilon} = \frac{P_{ET}}{(P_{ET} - P_E)} \tag{1}$$

$$Q_E \quad = \quad \frac{\eta_{TH}(M_N P_{neutrons} + P_\alpha + P_{IN} + \eta_{pump}P_{pump})}{(P_{aux} + P_{pump} + P_{IN}/\eta_{IN} + P_{cryo} + P_{sub} + P_{control})} \tag{2}$$

Pending detailed evaluation, the pump efficiency is taken to be $\eta_{pump} = 0.98$, the primary-coolant pumping power is $P_{pump} = f_{pump}\,P_{ET}$, and the auxiliary power is $P_{aux} = f_{aux}\,P_{ET}$, consisting of power to the coolant systems and housekeeping power. The ratio of fusion power to input power is the fusion gain, $Q \equiv P_F/P_{IN}$, where the fusion power is $P_F = P_{neutrons} + P_\alpha$. Reworking this equation to eliminate explicit powers, the required D-T fusion gain, Q, to achieve a target recirculating power ϵ derived from the power-flow diagram to be

$$Q \quad = \quad \frac{1}{(0.2 + 0.8M_n)}\left[\frac{(1/\eta_{TH} - f_{pump}\,\eta_{pump})(1/\eta_{IN})}{(\epsilon - f_{pump}\,\eta_{pump} - f_{aux})} - 1]\right]. \tag{3}$$

The above equation is applicable to most steady-state D-T fusion approaches. A design-space plot is provided as Fig. 3 of Ref. [14]. For systems with some recaptured (REC: direct conversion) power, the required gain, Q, becomes:

$$Q = \left[\frac{(1/\eta_{TH} - f_{pump}, \eta_{pump})(1/\eta_{IN} - \eta_{REC})}{(\epsilon - f_{pump}, \eta_{pump} - f_{aux})} - (1 - \eta_{REC})\right] \tag{4}$$

A design-space plot with $\eta_{REC} = 0.5$ is provided as Fig. 4 of Ref. [14].

For a pulsed fusion system, it is convenient to define the plasma internal energy as $W_{IN} = P_{IN}\tau_d$, where tau_d is the dwell time between fusion pulses. The fusion energy released per pulse is W_F such that the continuous fusion power available to the primary loop is $P_F = W_F/(\tau_d + \tau_b)$, where τ_b is the pulse (or burn) duration. Expressed in terms of energy, a gain, Q^*, can be defined as

$$Q^* = \frac{W_F}{W_{IN}} = \frac{P_F(\tau_d + \tau_b)}{P_{IN}\tau_d} = Q(1 + \tau_b/\tau_d) \tag{5}$$

A short τ_b implies a large fusion power pulse and Q^* approaching Q.

4 Power Accounting Table

Account	Power Account Description	Parameter	Power	Units
1	**Fusion power**			
1.1	Fusion Power	P_{fusion}	1290	MW
1.2	Alpha Power	P_{α}	258.29	MW
1.3	Neutron Power	$P_{neutrons}$	1031.71	MW
1.4	Neutron Energy Multiplier	$M_N = P_{TH}/P_{fusion}$	1.1	
1.5	P_{pump} capture efficiency	η_{pump}	0.5	
1.6	Thermal Power	P_{TH}	1286.76	MW
1.7	Thermal conversion efficiency	η_{TH}	0.47	
1.8	Total (Gross) Electric Power	P_{ET}	721.01	MW
1.9	Lost Power	P_{Lost}	694.9	MW
2	**Recirculating power**			
2.2	Coolant Pumping Power Frac.	f_{pump}	0.01	
2.2.1	Coolant Pumping Power	$P_{pump} = f_{pump} \cdot P_{ET}$	7.93	MW
2.3	Subsystem and Control Fraction	f_{sub}	0.01	
2.3.1	Subsystem and Control Power	$P_{sub} + P_{control} = f_{sub} \cdot P_{ET}$	3.61	MW
2.4	Auxiliary systems	$P_{aux} = P_t + P_h$	4.98	MW
2.4.1	Tritium Systems	P_t	2.84	MW
2.4.2	Housekeeping power	P_h	2.14	MW
2.5.5	Cryo vacuum pumping	P_{p_c}	1.62	MW
2.6	Input power	$P_{IN} = P_{comp} - P_{rec} + P_{tar}$	16.5	MW
2.6.1	Compression system efficiency	η_{comp}	0.25	
2.6.2	Power into compression system	P_{comp}	15	MW
2.6.4	Recovered power	P_{rec}	0	MW
2.6.5	Plasma target formation	P_{tar}	1.5	MW
2.6.6	Target formation efficiency	η_{tar}	0.25	
3	**Output power**			
3.1	Scientific Q	$Q = P_{fusion}/P_{IN}$	78.18	
3.2	Engineering Q	Q_E	8.56	
3.3	Recirculating power fraction	$\epsilon = 1/Q_E$	0.12	
3.4	Net Electric Power	$P_E = (1-\epsilon) \cdot P_{ET}$	636.75	MW

Table 1: Power balance for a pulsed magnetic fusion energy system using cyclic compression with some recovery of the compression energy.

5 Cost Category 10: Pre-construction costs

Cost Category 10, which encompasses Capitalized Pre-Construction Costs, is an essential aspect of project budgeting in plant construction. This category includes a range of preliminary expenses incurred before the actual construction begins. These costs cover the acquisition of land and associated rights, obtaining necessary permits and licenses for the site and plant, and conducting essential studies and reports to ensure compliance and feasibility. Additional elements include various other pre-construction expenditures and a contingency allocation to account for unforeseen costs in these areas. This category is fundamental in laying the groundwork for a successful construction project, ensuring that all legal, environmental, and logistical bases are covered in preparation for the actual building phase. Total costs for for Cost Category 10 are $ 42 M.

Cost Category - 11 Land and Land Rights

This Cost Category includes the purchase of new land for the reactor site and land needed for any co-located facilities such as dedicated fuel cycle facilities. Costs for acquisition of land rights should be included.

2020 update *Scope:* Purchase of new land *Previous values:* Different acreage estimates across the four fusion plant concepts based on their fusion power rating. Cost of \$10,900/acre. Prior example: 462 acres $\times$ \$10,900/acre = \$4.9 million / 150 MW plant capacity = \$33/kW. Min \$14/kW, Average \$20/kW, Max \$33/kW *Reduction strategies:* Use of existing power plant site (potentially streamlined siting process if an existing nuclear fission plant site, perhaps after the fission plant has retired, but could also be a coal plant for repowering with fusion); minimum plant site acreage (much less need for buffer area than fission plant because fusion plants ; minimum price acre (marginal land far from cities (although it could also be argued that low radiation source terms will allow siting closer to cities)). Representative plant acreage and cost per acre for greenfield plant site: 400 acres $\times$ \$10,000/acre = \$4 million / 150 MW plant capacity = \$27/kW, giving 23.8 M \$.

2022 update A fusion power core will have a footprint as discussed in Cost Category 21, consisting of buildings that contain the fusion power core, turbine hall and hotcell. For most concepts, the site will also house switchyards and heat rejection (cooling towers). The site size will be set by the regulating authority, which will prescribe a site size proportional to the tritium inventory. The cost basis here is to determine the costs of the site beyond the boundary of the buildings scale in linear proportion to neutron power, with aneutronic fuels requiring the smallest boundary. Note also that these costs will not be required for retrofitting an existing power plant, or for heat sources that are brought onto an existing site. Greenfield site costs will vary by location. The costs are 9 M for a system comprising of 1 modules. The expression used to calculate the cost is given below:

```
C_20 = sqrt(N_mod) * (P_NEUTRON /239 * 0.9 + P_FUSION/239*0.9)
```

Cost Category 12 – Site Permits

This Cost Category includes costs associated with obtaining all site related permits for subsequent construction of the permanent plant. The total for this element is \$ 10 M.

Cost Category 13 – Plant Licensing

This Cost Category includes costs associated with obtaining plant licenses for construction and operation of the plant, typically \$1 M to \$10 M. This range accounts for the technical and engineering studies, safety analyses, and environmental impact assessments required. The total for this element is \$ 10 M.

Cost Category 14 – Plant Permits

This Cost Category includes costs associated with obtaining all permits for construction and operation of the plant, typically \$ 500,000 to \$ 5M. This cost can escalate if the licensing process is prolonged or if there are unique legal challenges. The total for this element is \$ 5 M.

Cost Category 15 – Plant Studies

This Cost Category includes costs associated with plant studies performed for the site or plant in support of construction and operation of the plant. For new designs, this can range from \$ 10 M to over \$ 100 M, especially if new safety features or innovative technologies are being developed. The total for this element is \$ 5 M.

Cost Category 16 – Plant Reports

This Cost Category includes costs associated with production of major reports such as an environmental impact statement or the safety analysis report, usually in the range of \$ 500,000 to \$ 5 M. Comprehensive site evaluations, especially in environmentally sensitive areas, can be expensive. The total for this element is \$ 2 M.

Cost Category 17 – Other Pre-Construction Costs

This Cost Category includes other costs that are incurred by Owner prior to start of construction and may include public awareness programs, site remediation work for plant licensing, etc. Typically, \$ 100,000 to \$1 M. Costs depend on the extent of the consultation process and the public's level of interest and concern. The total for this element is \$ 1 M.

Cost Category 19 – Contingency on Pre-Construction Costs

This Cost Category includes an assessment of additional cost necessary to achieve the desired confidence level for the pre-construction costs not to be exceeded. We have set the basis as 10 % of the total costs in this major category. The total for this element is therefore \$ 0 M.

6 Cost Category 20: Capitalized Direct Costs (CDC)

We borrow all of the two-digit cost accounts from the GENIV EMWG Guidelines, which depart slightly from the IAEA 2001 guidelines. Each subcategory is broken out separately. In the EMWG Guidelines document, cost category 20 is designated as "Capitalized Direct Costs" (CDC). This category is a key component of the overall cost structure for energy plant projects and includes various essential expenditures directly related to the construction and commissioning of a plant. Typically, such costs encompass all direct expenses incurred in the physical construction of the plant, including materials, labor, and any other resources directly associated with the construction activities.

6.1 Cost Category 21: Structures and Improvements

This account covers all direct costs associated with the construction and provision of physical plant buildings and structures. Key elements include the reactor, turbine, electrical equipment, and cooling system structures, along with site improvements, facilities, and miscellaneous building work. The costs for these structures, especially the heat island and turbine buildings, represent a significant portion of the total cost of the facility. For instance, the Heat Island Building is noted for its compact design, thick radiation shielding walls, and inclusion of access corridors and hot cell, leading to its estimated cost. The Turbine Building, housing turbines and auxiliary equipment, is also a major cost component, with its size and cost dependent on the size of turbines and equipment. Additional facilities like the Hot Cell, Service Water, and Fuel Handling Buildings, Control Room, On-Site AC Power, and other administrative and service buildings are also included, each having distinct scaling factors and cost estimates based on their specific functions and requirements within the power plant. We provide a detailed cost estimation approach for each of these buildings, reflecting their critical role in the overall construction and operation of the energy plant.

2020 Update *Scope:* Site preparation and yard work, reactor building, turbine building, security building and gatehouse, control and administrative buildings, maintenance shop, any other site structures and facilities. *Previous values:* CAD calculations for building volumes and concrete volumes multiplied by materials costs. Min \$373/kW, Average \$606/kW, Max \$1,142/kW. *Reduction strategies:* Minimal reactor building size and wall/ceiling thickness (while fully preserving safety); minimal volumes for concrete and other materials (expensive concrete vs. inexpensive soil or sand, perhaps as inner fill material within metal or concrete walls); avoidance of unnecessary buildings in plant design; reuse of buildings if an existing power plant site; modular construction, as in Wartsila Modular Block. *Updated value:* Average of previous value reduced by 25 % with implementation of feasible strategies listed above. \$606/kW $\times$ - 0.25 = \$470/kW, giving a total for CAS21 of 258.1 M \$.

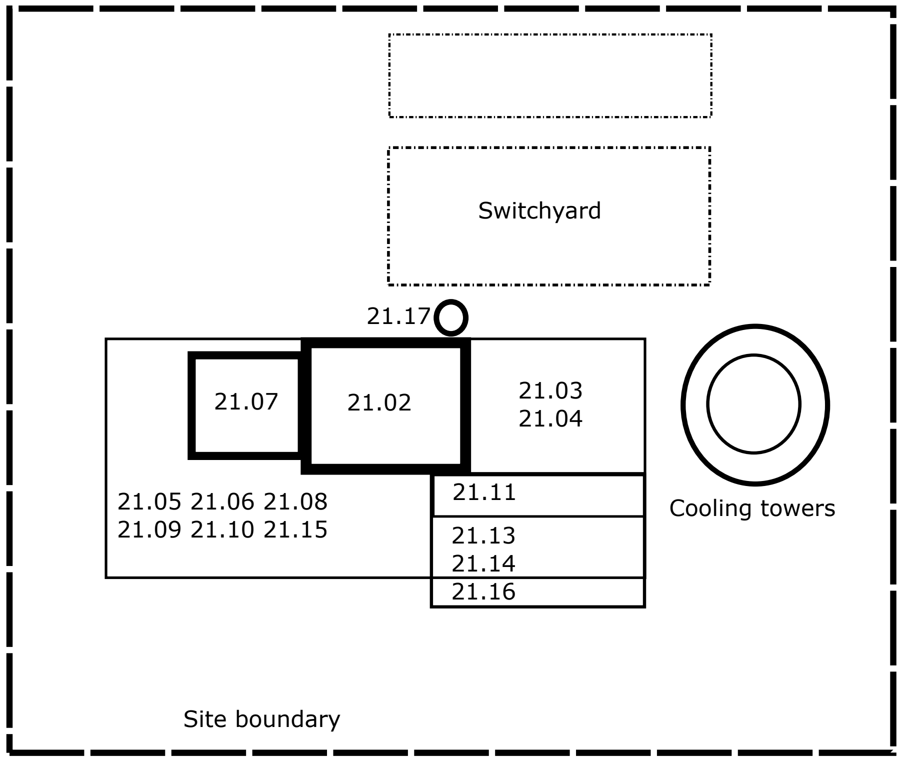

Figure 5: Site plan with labels according to the cost accounts described in the text. Cooling towers and Switchyards are costed elsewhere.

Cost Category	Building	Wall materials	$/kW gross	L	W	H	Vol	Esc year	Esc	$/kW gross
21.01.00	Site improve. & facs		20.7					2019	1.19	24.6
21.02.00	Heat Island Building	Concrete & Steel	131.6	48.3	48.3	60	140000	2009	1.42	186.8
21.03.00	Turbine building	Steel	45.3	48.3	48.3	30	70000	2019	1.19	54.0
21.04.00	Heat rejection	Concrete & Steel	31.7	48.3	48.3	15	35000	2019	1.19	37.8
21.05.00	Power supplies	Concrete & Steel	9.1	9.7	9.7	6.0	560	2019	1.19	10.8
21.06.00	Plant aux.	Concrete & Steel	4.5	4.8	4.8	3.0	70	2019	1.19	5.4
21.07.00	Hot cell	Concrete & Steel	65.8	24.2	24.2	60	35000	2013	1.42	93.4
21.08.00	Heat Island Services	Steel frame	13.2	4.8	4.8	10	233	2013	1.42	18.7
21.09.00	Service water	Steel frame	0.2	1.3	4.0	4.0	21	2019	1.19	0.3
21.10.00	Fuel storage	Steel frame	0.9	5.0	15.0	2.5	188	2019	1.19	1.1
21.11.00	Control room	Steel frame	0.7	4.0	12.0	2	96	2019	1.19	0.9
21.12.00	Onsite AC power	Steel frame	0.7	3.6	10.8	1.8	70	2019	1.19	0.8
21.13.00	Administration	Steel frame	3.7	20.0	60.0	10	12000	2019	1.19	4.4
21.14.00	Site services	Steel frame	1.3	7.3	22.0	3.7	593	2019	1.19	1.6
21.15.00	Cryogenics	Steel frame	2.0	11.0	33.0	5.5	2003	2019	1.19	2.4
21.16.00	Security	Steel frame	0.7	4.0	12.0	2	96	2019	1.19	0.9
21.17.00	Ventilation stack	Steel & concrete	22.7			120		2019	1.19	27.0
21.98.00	Spare parts allowance									0.0
21.99.00	Contingency allowance		0.0							0.0
21.00.00	Structures and site facs		354.9							470.7

Table 2: Cost categories for buildings, and their cost estimation

Cost Category 21.1 Site Preparation/Yard Work

Includes clearing, grubbing, scraping, geo-technical work, site cut, fill and compact, drainage, fences, landscaping, etc. Cost is $ 15.3 M using NETL Reference case B12A, account 13, bare erected costs for 13.1 site preparation and 13.2 site improvements, for a 686MW gross power plant in 2019 $, applying escalation per table 2.

Cost Category 21.2 Heat Island Building

Includes installation, labor, and materials for concrete and metalwork for the building surrounding and supporting the heat island, including the containment structure. Also includes the biological shielding, structural excavation and backfill, foundations, walls, slabs, siding, roof, architectural finishes, elevators, lighting, HVAC (general building service), fire protection, plumbing, and drainage. Cost is $ 95.1 M using

Waganer cost reference for ARIES-ST.

Cost Category 21.3 Turbine Generator Building

Includes installation, labor, and materials for concrete and structural metalwork for the building surrounding and supporting the turbine generator(s). (For concepts that do not produce electricity, this account can be replaced with appropriate energy product buildings.) Also includes structural excavation and backfill, foundations, walls, slabs, siding, roof, architectural finishes, elevators, lighting, HVAC, fire protection, plumbing, and drainage. This building contains the turbines and heat exchangers. Cost is $ 64.1 M using NETL Reference case B12A, account 14.3, turbine building bare erected costs for a 686MW gross power plant in 2019 $, apply escalation per table 2.

The rest of the 21 series accounts are for other support buildings on the site. Modular concepts might have a separate building to house centralized functions for all modules, such as an external control room. Here, the building costs are for the complete civil structure, including structural excavation and backfill, foundations, finishes, and building services such as elevators, lighting, HVAC, fire protection, or domestic water and drainage, but do not include the specialized equipment within.

- Cost Category 21.04 - Heat Rejection Building. This account covers the costs associated with the systems needed for rejecting heat from the power plant, scaled to the rejected thermal power. Cost is $ 12.8 M using NETL Reference case B12A, account 14.2, boiler building bare erected costs for a 686MW gross power plant in 2019 $, apply escalation per table 2.
- Cost Category 21.05 - Electrical Equipment & Power Systems Building. It includes the costs for all electrical equipment and power systems, scaling with the net electrical power output of the plant. Cost is $ 20.4 M, using a smaller version of the turbine building - multiple floors, concrete, per table 2.
- Cost Category 21.06 - Plant Auxiliary Systems Building. This account addresses the costs of auxiliary systems in the plant, such as compressed air, inert gas storage, and distribution systems, scaling with the gross electrical power. Cost is $ 1.0 M as smaller version of the turbine building - multiple floors, concrete, per table 2.
- Cost Category 21.07 - Hot Cell Building. The cost for the Hot Cell Building, designed for handling large and hazardous components, is scaled in relation to the Reactor Building. Cost is $ 21.9 M as a smaller version of the Heat Island Building by 50%, per table 2.
- Cost Category 21.08 - Heat Island Service Building. This includes the costs for a service building specific to the reactor, similar in cost to structures in projects. Cost is $ 2.2 M as smaller version of the Heat Island Building by 10%, per table 2.
- Cost Category 21.09 - Service Water Building. The Service Water Building, essential for the plant's water supply needs, has its costs modeled after similar structures in other power plant projects. Cost is $ 1.0 M following NETL Reference case B12A, account 14.5, circulation water pumphouse and 14.6 water treatment building bare erected costs for a 686MW gross power plant in 2019 $, applying escalation per table 2.
- Cost Category 21.10 - Fuel Handling and Storage Building. This account covers the costs for buildings related to handling and storing fuel, scaled according to tritium usage or fusion power. Cost is $ 2.2 M, scaled relative to the administration building per table 2.
- Cost Category 21.11 - Control Room Building. It encompasses the expenses for the control room building, essential for plant operations, with costs similar to those in previous plant designs. Cost is $ 3.1 M scaled relative to the administration building per table 2.
- Cost Category 21.12 - On-Site AC Power Building. This account includes the costs for buildings housing on-site alternating current (AC) power systems, comparable to similar structures in other projects. Cost is $ 2.9 M scaled relative to the administration building per table 2.

- Cost Category 21.13 - Administrative Building. It covers the costs associated with the construction of the plant's administrative building, with cost estimates based on previous similar projects. Cost is \$ 3.1 M as NETL Reference case B12A, account 14.4, administration building building bare erected costs for a 686MW gross power plant in 2019 \$, apply escalation per table 2.

- Cost Category 21.14 - Site Service Building. This account entails the costs for a building providing various site services, with costs aligned with those in comparable plant designs. Cost is \$ 2.0 M as NETL Reference case B12A, account 14.7, machine shop building bare erected costs for a 686MW gross power plant in 2019 \$, apply escalation per table 2.

- Cost Category 21.15 - Cryogenic and Inert Gas Storage Building. It includes the costs for buildings that store cryogenic and inert gases, essential for plant operations. Cost is \$ 2.9 M Like site services, per table 2.

- Cost Category 21.16 - Security Building. The Security Building account covers the costs for structures dedicated to the plant's security and surveillance. Cost is \$ 1.0 M like administration, see table 2.

- Cost Category 21.17 - Ventilation Stack. This account addresses the costs of the ventilation stack, a crucial component for plant ventilation and safety. Cost is \$ 7.1 M as NETL Reference case B12A, account 7, ventilation stack bare erected costs for a 686MW gross power plant in 2019\$, apply escalation per table 2.

6.2 Cost Category 22: Heat Island Plant Equipment

This category is most dependent on the heat island technology being considered, because the sub-account descriptions and costs depend heavily on the coolant used and whether the subsystems are factory-produced or constructed onsite.

For the mechanical components, costs are determined by making a CAD drawing of the components and determining the volume of the component. The following equation is then used to determine component costs from the volumes: $\Sigma_i C_i = V_i \rho_i c_i$, where ρ is the mass density in kg/m^3 and c_i is the cost per unit mass in \$/kg (see Table 3) with manufacturing factor applied.

Material	ρ (kg/m^3)	c_{raw} (\$/kg or \$/kAm)	Manufacturing Factor	σ (MPa)	c (\$/kg or \$/kAm)
FS	7470	10	3	450	
Pb	9400	2.4	1.5		
Li_4SiO_4	2390	1	2		
FLiBe	1900				40
W	19300	100	3		
Li	534	70	1.5		
BFS	7800	30	2		
PbLi	8594.0				12.82
SiC	3200	14.49	3		
Inconel	8440	46	3		
Cu	7300	10.2	3		
Polyimide	1430	100	3		
YBCO	6200				55
Concrete	2300	0.52	2		
SS316	7860	2	2	900	
Nb_3Sn					5
Incoloy	8170	4	2		
GdBCO					
He					
NbTi					

Table 3: Properties of materials used in fusion reactors.

6.2.1 Cost Category 22.01: Heat Island Components

The fusion system comprises various critical components, each playing a unique role in ensuring the effective and safe operation of the fusion process. These components are:

- Cost Category 22.01.01 First Wall and Blanket. The first wall and blanket are crucial for protecting the reactor's interior from intense fusion reactions and for capturing and converting fusion energy into heat.
- Cost Category 22.01.02 Shield. The shield serves to protect surrounding structures and personnel from radiation produced during fusion, playing a key role in ensuring safety and regulatory compliance.
- Cost Category 22.01.03 Magnets. In magnetic fusion systems, magnetic field coils are used to confine and control the fusion reaction, and provide isolation from the surfaces.
- Cost Category 22.01.04 Compression system. This includes compression systems that assist in raising the plasma pressure to the required level for fusion reactions to occur.
- Cost Category 22.01.05 Primary Structure and Support. The primary structure and support systems provide the necessary mechanical integrity and stability to the fusion reactor.
- Cost Category 22.01.06 Vacuum System. The vacuum system maintains the necessary vacuum conditions within the reactor chamber, which is crucial for the containment and control of the plasma.
- Cost Category 22.01.07 Power Supplies. These are essential for providing the electrical power needed for various reactor systems, including heating, cooling, and plasma confinement.
- Cost Category 22.01.08 Exhaust/divertor. In magnetic fusion systems, the exhaust gases are channeled to a surface called a divertor.
- Cost Category 22.01.09 Direct energy convertor. In some cases, energy can be directly converted from the exhaust plasma to electricity with higher conversion efficiency.
- Cost Category 22.01.11 Assembly and Installation. This encompasses the human resources involved in the assembly and installation of the fusion heat island components.
- Cost Category 22.01.19 Scheduled replacement costs. Capitalized costs of components that will not serve the lifetime of the plant and be replaced on a schedule - usually the first wall and blanket, but can be components in other systems listed here.

Each of these major costs are examined in detail in the following sections.

Cost Category 22.01.01 First Wall and Blanket

The first wall in a fusion heat island serves as a direct interface with the plasma, tasked with withstanding extreme heat and particle flux while also resisting radiation damage and managing tritium retention. The blanket surrounding the first wall plays a dual role in absorbing neutrons to generate heat and breed tritium for fuel, in addition to providing crucial radiation shielding for the reactor's structural integrity and safety. This Cost Category therefore consists of the following subcategories:

- Cost Category 21.01.01 First wall: the interior surface of the fusion chamber that directly faces the plasma and is subject to extreme conditions. It endures intense heat and particle flux, necessitating the use of materials like refractory metals or specialized alloys that can withstand high temperatures and minimize erosion from particle bombardment. The first wall must also resist radiation damage, particularly in deuterium-tritium fusion reactors, where neutron radiation can cause materials to become brittle and lose structural integrity. An important consideration is the management of tritium retention and release, as wall materials should prevent buildup while allowing efficient extraction and reuse. Additionally, the first wall faces significant thermal stress due to the cyclic nature of fusion reactions, requiring a design that can accommodate stresses without cracking or deforming. Finally, it must be compatible with the cooling system, which could involve water, liquid metal, or other coolants, to ensure effective heat removal. The design and performance of the first wall are critical for the reactor's efficiency, safety, and longevity, which presents a complex engineering challenge that balances heat resistance, structural integrity, and minimal plasma interaction.

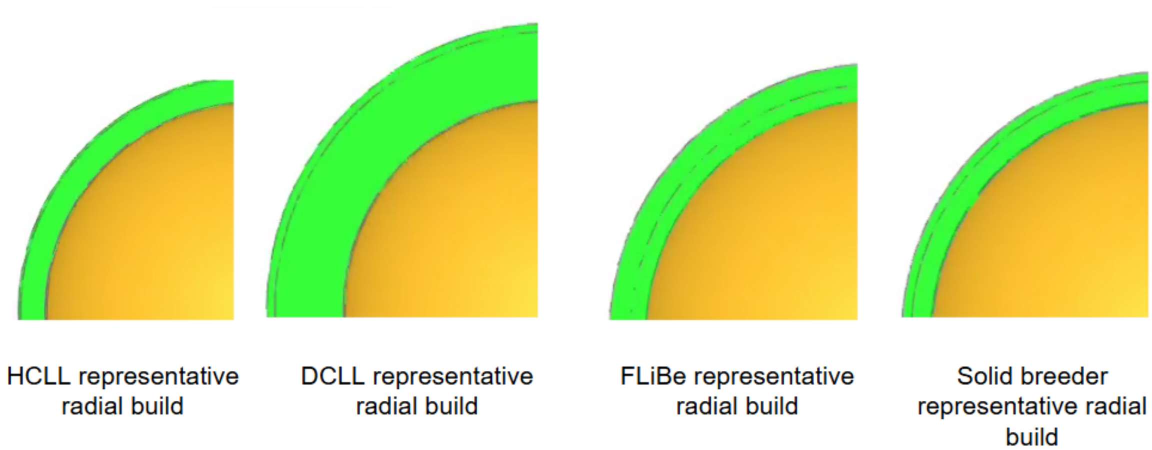

Figure 6: Blankets that we considered before downselecting to the DCLL.

- Cost Category 21.01.02 Blanket: a crucial component that surrounds the plasma, serving multiple roles. It absorbs neutrons generated during fusion reactions, converting their kinetic energy into heat, and acts as a breeding ground for tritium, an essential fuel in deuterium-tritium fusion reactions, through interactions with materials like lithium. The blanket also provides critical radiation shielding, protecting reactor components and personnel from intense neutron radiation. Additionally, it plays a role in thermal management, as the heat generated within the blanket is often harnessed to produce steam for driving turbines and generating electricity. This makes the design and material choice for the blanket vital for reactor efficiency and safety, which requires advanced materials and engineering to withstand extreme temperatures and radiation levels.

The blanket under consideration has a tungsten first wall with a primary coolant consisting of Lead Lithium (PbLi) and water as the secondary coolant, with Pb as part of PbLi neutron multiplier, and a structure primarily consisting of Ferritic Martensitic Steel (FMS). The radial build dimensions are shown in Table 4, which allow us to determine the volumes of components for costing. The radial build is shown in 7.

	Thickness	Inner Radius	Outer radius	Volume
Plasma	1.1	0	1.1	57.0 m^3
Vacuum	0.1	1.1	1.2	10.8 m^3
First Wall	0.2	1.2	1.4	24.5 m^3
Blanket 1	0.8	1.4	2.2	135.7 m^3
Structure	0.2	2.2	2.4	43.4 m^3
Reflector	0.2	2.4	2.6	47.1 m^3
Blanket 2	0.5	2.6	3.1	134.3 m^3
Vessel	0.5	3.1	3.6	157.9 m^3
Shield	0.5	3.6	4.1	181.4 m^3
Bioshield	1.0	4.1	5.1	433.5 m^3

Table 4: Volumes of components in the radial build.

The costs of the first wall and blanket are determined by the volume of the material and multiplied by a manufacturing factor per Table 3. Cost of the first wall is \$ 140.0 M. Cost of the blanket is \$ 7.6 M.

Cost Category 22.01.02: Shield

The shield comes in 4 major components - the high temperature shield, the low temperature shield, the bioshield and the penetration shield. Each shield serves a vital role in ensuring both the safety and functionality of the heat island. Primarily, they act as barriers to protect outer components and personnel from

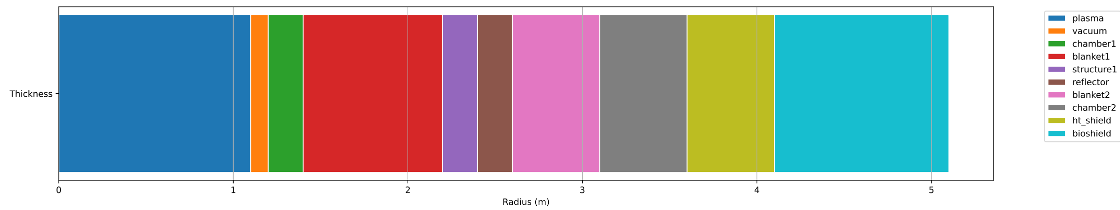

Figure 7: Radial build according to Table 4.

the intense neutron radiation and heat generated by the fusion reactions. The shields effectively absorbs and attenuates this radiation, preventing it from causing damage or activating surrounding materials. Additionally, the shields help to maintain the structural integrity of the heat island components and may also play a role in thermal management, capturing and dissipating residual heat.

This Cost Category comprises:

- Cost Category 22.01.02.01 High temperature shield consists of a steel structure that envelopes the plasma and blanket system with holes running along the length of the shield for Lead Lithium (PbLi) coolant to flow through, most often placed inside of the vacuum vessel. The basis is determined by the volume of material shown in Table 4, giving a volume of 181 m^3 multiplied by a manufacturing factor, to obtain a cost of $ 78 M.
- Cost Category 22.01.02.02 Low temperature shielding, which is placed outside of the vacuum vessel and is not cooled. This structure is usually steel. This component is not included in the design hence its cost is $ 0 M.
- Cost Category 22.01.02.03 Bioshield, typically made of dense, neutron-absorbing materials like concrete, lead, or borated materials, the shield is a crucial component for the safe and efficient operation of a fusion energy system, enabling it to harness the power of fusion while minimizing the associated risks. The bioshield is computed also from volumes in Table 4, giving a volume of 181 m^3 and a cost of $ 2 M.
- Cost Category 22.01.02.04 Penetration shielding consists of the doglegs that are put in place to allow entry or cabling, conduits or coolant systems into the bioshield. These are not used in this design hence their cost is set to $ 0 M.

The total cost for the shield subsystems is then $ 80 M.

Cost Category 22.01.03: Coils

There are no coils, except in the target injetor - those are considered in 22.1.8.

Cost Category 22.01.04: Compression System

The LANL concept uses coaxial plasma guns to accelerate a plasma to 50km/s, which become spherically (or obloidally) convergent on a target plasma. The kinetic energy of the jets compress a plasma to high density and fusion temperature. The fusion energy design involves the dynamic compression of a magnetically-confined plasma by a gaseous liner.

The major components of the LANL concept are therefore:

- Cost Category 22.01.04.01 Plasma Guns. It is anticipated that to obtain the required symmetry of implosion up to 300 guns will be needed. The plasma is accelerated along a 1m coaxial unmagnetized plasma gun, made of copper and tungsten. The aperture radius is 0.2m. Each gun is estimated to cost 5000 dollars, costing in total $ 2 M.

- Cost Category 22.01.04.02 Coaxial transmission line. In order to get sufficient shielding between the chamber and the capacitor banks, and to shield the outside world from neutrons, a low inductance coaxial transmission line will be needed for each gun, with a minimum length of 1m (likely closer to 2m to accommodate all of the capacitors that will be needed to energize the plasma guns). This cost is $ 1 M.
- Cost Category 22.01.04.03 Fast valves to allow for separation of the gun and chamber for clearing of the chamber and for maintenance. Valves will cost $ 1 M.
- Cost Category 22.01.04.04 Diagnostic and Control System. Responsible for monitoring and controlling the entire setup, this system collects real-time data and adjusts the other components to maintain desired fusion conditions and ensure operational safety. The cost is $ 3 M.
- Cost Category 22.01.04.05 Coolant System. Responsible for maintaining optimal operational temperature of the guns. This cost scales with energy of the bank and shot frequency, assuming that 3/5 of the losses go to heat. The cost is $ 45 M.

Power supplies for the plasma guns are treated in 22.01.07. The cost of this element therefore totals $ 25 M.

Cost Category 22.01.05: Primary Structure and Support

This account defines the cost of the primary structure and support for the power core. This subsystem transfers the gravity and seismic loads to the building support structures. Elements include reactor pressure vessel supports, brackets, sealings, pipe supports, or others, including shielding materials if they are integral parts of the support structure.

A key requirement of the primary support structure is to absorb and mitigate seismic loads. When building nuclear fission reactors, the accounting of seismic events has been a key cost driver. This is because near-surface soils and seismic hazard are different at each site, requiring site-specific analysis, design, engineering, qualification, licensing, and regulatory review, essentially making every design First-of-a-Kind (FoaK). Thus, a standardized approach is required to mitigate this cost for wide-scale, multiregion fusion reactor deployment.

The cost of this category has been largely based on a survey of the structural costs of nuclear fission conducted by Lal et al. [15]. Costs associated with this system can be broken into the following.

- 22.02.05.01 Engineering costs. This includes . The cost to analyze/design for operational loadings and calculate its seismic capacity, the cost to seismically qualify the first unit, the cost to prepare for, and of, regulatory review.
- 22.02.05.02 Fabrication costs. This includes the cost to fabricate the first unit, including materials, tooling, etc., the cost to fabricate the tenth unit identical to the first unit, the cost to increase seismic capacity of the first unit for the specified peak ground acceleration (PGA) required for the region. This includes all costs of analysis, design, seismic qualification, and peak ground acceleration regulatory review, and the cost to fabricate the first unit with seismic capacity associated with the required PGA.

It has been shown [15] that the costs for this category are significantly impacted by the seismic load, and decreasing the PGA at the reactor, greatly reduces costs. One means for isolation is to mount the reactor on numerous concave Friction Pendulum (FP) bearings. By employing these FP bearings, a cost saving of 40-50% can be achieved [15].

The average costs for a FOAK reactor from 11 facilities can be found for each construction step for a 1000 MW reactor core.

To cost the system in this report, these costs are categorised under cost category 22.1.1: First Wall and Blanket (see 7), thus the final costs here are $ 0 M for engineering and $ 0 M for fabrication, totalling $ 0 M.

Cost Category 22.01.06: Vacuum System

The vacuum system in a nuclear fusion reactor plays a critical role in creating and maintaining a high-quality vacuum environment essential for plasma containment and fusion reactions. It comprises various components such as vacuum vessels, primary and roughing vacuum pumps, helium liquefier-refrigerators, and vacuum ducts, each contributing to the effective evacuation and thermal management necessary for sustained and controlled fusion processes. Cost Category 22.1.6 Vacuum System consists of:

- 22.01.06.01 Vacuum Vessel. The vacuum vessel is a sealed container designed to maintain a high vacuum environment, essential for minimizing interactions between the plasma and any residual gases, thus enabling controlled fusion reactions.
- 22.01.06.02 Helium Liquefier-Refrigerators. Helium liquefier-refrigerators are used to cool components of the reactor, such as superconducting magnets, to extremely low temperatures, often necessary for maintaining superconductivity and efficient thermal management.
- 22.01.06.03 Primary Vacuum Pumps. Primary vacuum pumps are responsible for creating and maintaining the vacuum within the vessel by removing gases and air, which is crucial for the initial establishment of the vacuum environment.
- 22.01.06.04 Roughing or Backing Pumps. Roughing or backing pumps are used in conjunction with primary vacuum pumps, typically to provide the initial evacuation of the vessel and reduce the pressure to a level where primary pumps can operate effectively.
- 22.01.06.05 Vacuum Ducts. Vacuum ducts are the conduits that connect various components of the vacuum system, facilitating the efficient transport and removal of gases from the vacuum vessel to the pumps.

Each subsystem is considered in detail as follows.

22.01.06.01 Vacuum vessel: this account dominates the cost of the vacuum system and includes only the structural elements of the vacuum system, including the main vacuum vessel chamber, port enclosures, the vacuum door assembly, bulk shielding inside the walls of the vacuum vessel and any manifold / padding [16]. It serves two primary functions: (1) to maintain the vacuum environment created by the pumping system for the plasma experiments; (2) to provide the structural support for the superconducting magnets and the other internal and external systems. The design for this mirror concept is based on the vessel used for the MFTF-B at LLNL [17].

The vacuum vessel serves as a key structural element in the reactor, from which other systems including and cryogenic systems are mounted. As such, there are several key loading requirements:

- Gravity load. The vessel must support the 1 g force acting straight down on the entire vessel and the components it supports. The dominant masses are the vessel itself 101530 tonnes, the total coil mass COILM tonnes, and auxiliary components inside and outside the vessel, ∼500 tonnes.
- External pressure load. The external pressure load is caused by the standard atmospheric pressure acting on the vessel shell when the vessel is under vacuum, plus an additional 10% of the atmospheric load to account for magnetic pressure on the shell resulting from the sudden collapse of the magnetic field.
- Thermal cooldown load. The thermal cooldown load is caused by the magnets cooling to 4 K, while the vessel remains at room temperature. Because the magnets are connected to the vessel by a number of support struts on each magnet group, the possibility exists for an over-constrained strut system to generate thermal contraction loads.
- Environmental thermal load. The environmental thermal load is caused by the variation in the air temperature surrounding the vessel.

For the chamber geometry specified in Cost Category 22.01.02, a total material volume of 15 m^3 is calculated. ASTM A240 Type-304 stainless steel is used because it is nonmagnetic, has good fracture toughness and welding properties, is readily available and inexpensive, and a total mass of 101530 tons is required, costing 0.5 M USD. Considering the manufacturing and assembly process, however, will increase the vessel cost significantly. One notable requirement is to polish the entire inner surface to maintain the required high vacuum, and reduce the risk of electrical faults. An approximate manufacturing factor of 10 is assumed for the entire process. This results in a cost for vacuum vessel of 5 M USD.

Vessel volume (m^3)	1525
Material volume (m^3)	15
Material	ASTM A240 304 SS
Material mass (tons)	101530
Material cost (M USD)	0.5
Total cost (M USD)	5

Table 5: Vacuum vessel parameters.

22.01.06.02 Helium Liquefier-Refrigerators: This subsystem is the helium refrigeration and liquefaction process equipment. It is responsible for supplying the cooling necessary to maintain an operation temperature of 20 K at coils, absent in this case.

The costs for this subsystem are 5 M USD for the vacuum vessel, 0 M USD for the cooling systems, 15 M USD for the vacuum pumping systems, and 0.34 M USD for the roughing pump. The total cost is 20 M USD.

22.01.06.03 Primary Vacuum Pumps: consists of d a distributed vacuum subsystem with vacuum pumps located at each sector. For a vacuum of 1525 m^3, novpumps vacuum pumps will be required, costing 50 K USD each, corresponding to a total pumping system cost of 15 M USD.

22.01.06.04 Roughing Pumps: Consists of a single roughing pump to establish an initial vacuum and will cost 0.34 M USD.

Cost Category 22.01.07: Power Supplies

This category covers an extensive range of power supply systems critical for operation. This includes High-Voltage Power Supplies for plasma heating, Pulsed Power Supply Systems for plasma initiation, and Superconducting Coil Power Supplies for magnetic confinement. Auxiliary Heating Power Supplies support additional plasma heating, while Cooling and Cryogenic Systems maintain optimal temperatures for superconducting components.

- Cost Category 22.07.01 High-Voltage Power Supplies. Essential for driving heating systems like neutral beam injectors, which heat the plasma to the necessary temperatures for fusion.
- Cost Category 22.07.02 Pulsed Power Supply Systems. These provide high-power electrical pulses required for plasma initiation and shaping. This is the dominant cost category here. We have built 0.5MJ, 20kV, 500kA capacitor banks for 200k USD each, we multiply by 15 to get to the 6MJ needed for each pulse, and operate at 5kV to get lifetime needed for life of plant (see Fig. 8). We further multiply the number of banks we need by 5 to get the charge timing. Cost of this element is $ C220702 M.
- Cost Category 22.07.03 Coil Power Supplies: These are crucial for large superconducting magnets, and copper coils for equilibrium control.
- Cost Category 22.07.04 Auxiliary Heating Power Supplies. These are used for additional plasma heating systems like ion cyclotron resonance heating (ICRH) and electron cyclotron resonance heating (ECRH).
- Cost Category 22.07.05 Cooling and Cryogenic Systems. To maintain the necessary low temperatures for the superconducting magnets and other components.

- Cost Category 22.07.06 Control and Instrumentation Power Supplies. For the myriad of sensors, diagnostics, and control systems involved in monitoring and controlling the fusion reactor.
- Cost Category 22.07.07 Power Conversion Systems for Plasma Heating. Such as gyrotrons for ECRH and klystrons or tetrodes for ICRH.
- Cost Category 22.07.08 Safety and Interlock Systems. To ensure safe operations, especially during abnormal events or emergencies, these systems can cut power or initiate shutdown sequences.
- Cost Category 22.07.09 Diagnostics and Monitoring Systems Power Supply. For real-time monitoring of plasma and reactor conditions, requiring reliable and clean power sources.

This system is costed in two main parts: a calculation of the power requirements and cost based on the details of the confinement system, and a top-down costing for the full electrical power supply and distribution system, scaled from ITER. For this external infrastructure required for ITER, a 500 MW fusion system, the total cost was 269.6kIUA or 539M USD. We expect these FOAK costs to follow a usual learning curve for NOAK, and so apply an 80% learning curve credit to the ITER cost. Thus, with the bottom-up component for the compression system, and the top-down component for external infrastructure, the total cost is found: for compression system power supplies, the cost is $ 75 M, and $ 8 M for the full electrical power supply and distribution system.

The total cost is therefore 111 M USD for the 1290 MW fusion system presented here.

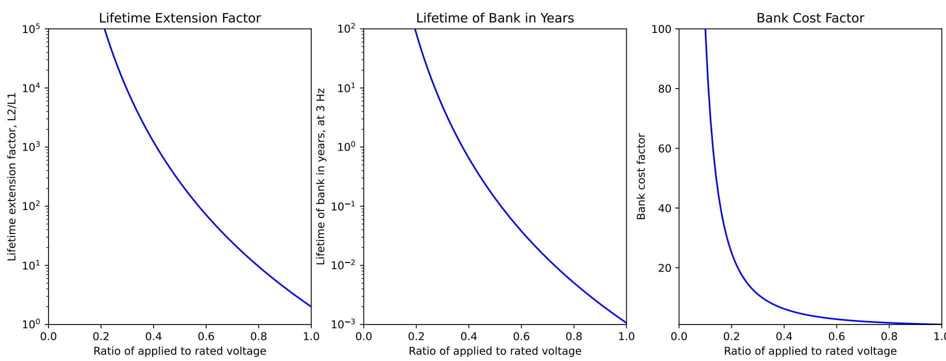

Figure 8: Lifetime extension factor vs ratio of applied to rated voltage.

Cost Category 22.01.08 Plasma Injector

Cost Category 22.01.08 FRC injectors

The target injection system is based on prior work by Grabowski.

Consists of:

- Cost Category 22.01.08.01 Theta pinch coils. This subsystem deals with the materials used in the divertor's construction, particularly those that can withstand high heat and neutron flux, such as tungsten or carbon-based composites.
- Cost Category 22.01.08.02 Buswork. Connecting the theta pinch coil to the capacitor bank via a low inductance and low resistance conducting connector.
- Cost Category 22.01.08.03 Vacuum system. Separate system to pump out the injector.
- Cost Category 22.01.08.04 Cooling. Theta pinch coil is copper and therefore lossy, and heat will need to be pulled out of the coils continuously.

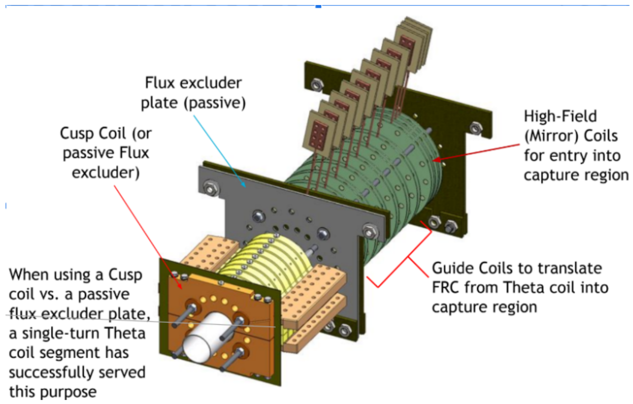

Figure 9: Caption

- Cost Category 22.01.08.05 Diagnostic and Monitoring Subsystem. This includes sensors and diagnostic tools that monitor the performance of the injector, including temperature, plasma density, and impurity levels. This subsystem is critical for injector operation control and for informing maintenance and operational adjustments.

The total cost is 10 M USD.

Cost Category 22.01.09: Direct Energy Convertor

A direct energy converter for fusion applications is a groundbreaking advancement in energy technology, notable for its remarkable efficiency of up to 70% in converting plasma energy to electricity. Key components include the Expander Tank, which reduces the power loading from diverted plasma, and the Expander Coils, essential for directing plasma towards electrodes that convert the energy. Consists of:

- Cost Category 22.01.09.01 Expander Tank. Converts surface power loading to tolerable levels from the diverted plasma to incidents surfaces.
- Cost Category 22.01.09.02 Expander Coils. Provides the magnetic funnel to direct plasma towards surfaces.
- Cost Category 22.01.09.03 Neutron Trap. Provides shielding of components in the direct energy converter from neutron damage.
- Cost Category 22.01.09.04 Vacuum System (separate from the chamber vacuum system). Provides pumping on the direct energy converter.
- Cost Category 22.01.09.05 Collector Electrode system. Allows charged particles to be captured on electrode surfaces for energy to be directly converted.
- Cost Category 22.01.09.06 Thermal Management. Cooling system for electrodes - either passively radiatively or actively with a liquid or gaseos coolant. Includes heat rejection system.
- Cost Category 22.01.09.07 Electrical subsystem. Provides the coupling to the balance of plant systems, listed in Cost Category 24. Electric Plant Equipment.

The collector can accrue damage from sputtering caused by the ion fluence. For tungsten ribbons, this erosion rate is estimated to be 0.02 mm/year, resulting in a low likelihood of having to replace grid elements.

Using additively manufactured ribbons with a simple water coolant channel, a heat flux of 2 MW/m^2 is acceptable, reducing the overall volume and thus the cost of the subsystem by a factor of 60% compared

Item	Cost (M USD)
Expander tank	5.7
Expander Coil and Neutron Trap Coil	11.7
Convertor gate valve	0.0
Neutron Trap Shielding	0.4
Vacuum system	5.7
Grid system	9.5
Heat collection system	2.1
Electrical	4.6
Cost per unit	39.6
Total unit cost	158.0
Engineering (15%)	23.7
Contingency (15%)	0.0
Total facility cost	209.0

Table 6: Costs for the direct energy convertor subsystems.

with a 1 MW/m^2 flux radiatively-cooled system.

An embodiment of a direct energy converter consists broadly of an expander and collector, comprising a grid of ribbons arranged in a 'venetian blind' configuration [18]. Various expander geometries have been proposed [18]. A conical configuration is often selected, due to the lower cost below 800 keV, as well as the ability to maintain the individual modules [19]. This direct energy conversion has an efficiency of $\sim$ 60 - 70% [20]. Thus, with the converter structure, the total cost is $ 0 M.

Cost Category 22.01.11: Assembly and Installation Costs

Consists of: assembly and installation costs of components of the heat island from the first wall through to blanket and shield. Modular installation is assumed, namely parts are assembled into modules at a centralized manufacturing location and are shipped for installation. Costs are broken out by major subsystem as follows.

- Cost Category 22.01.11.01 First wall and blanket
- Cost Category 22.01.11.02 Shield
- Cost Category 22.01.11.03 Magnets
- Cost Category 22.01.11.04 Auxiliary Heating
- Cost Category 22.01.11.05 Primary structure
- Cost Category 22.01.11.06 Vacuum system
- Cost Category 22.01.11.07 Power supplies
- Cost Category 22.01.11.08 Divertor
- Cost Category 22.01.11.09 Direct energy convertor

The total construction time of 0.5 is assumed for each subsystem, with 0.0 technicians, each billing 0.0 per day. Total labor cost of the installation of all heat island subsystems is 4.8 M USD.

Cost Category 22.01.19 Scheduled Replacement Cost

A key component of Cost Category 22.01 is the *Scheduled Replacement Cost*. This cost was previously included in the annualized costs when computing the LCOE, however, the LCOE annualized Cost Categories (70, 80 and 90) now have rules that exclude capitalized costs - the components in the heat island will be capitalized as they will be purchased at the time of construction of the heat island so they are ready for installation on a schedule to replace components before failure. This cost involves:

1. Component Replacement. Regularly scheduled replacement of components, which is essential for maintaining operational efficiency and safety. Only some components are ever considered 'life of plant', and this may well evolve to be all components in some fusion heat islands, however, very likely all components in Cost Category 22 will require scheduled replacement for the FOAK.

2. Replacement Schedule. Components such as those in the first wall and the blanket are replaced every replacementT years. Obviously for a FOAK power plant the replacement schedule may be must faster than for a NOAK system, and so this will be a major cost driver in systems that require regular replacement.

The annual replacement cost is calculated to be $40M, ensuring maximum availability.

6.2.2 Cost Category 22.02: Main and Secondary Coolant

This cost category is focused on the coolant systems in the fusion heat island. It encompasses the primary coolant system, including Lead Lithium (PbLi) as the primary coolant, along with its essential components such as pumps and motor drives, pipes, heat exchangers, various tanks (including dump, make-up, clean-up, tritium, and hot storage tanks), a clean-up system, thermal insulation for piping and equipment, tritium extraction, and a pressurizer, all shown schematically in Fig. 10. Similarly, it covers the secondary coolant system, which uses water and has comparable subsystems. The category also includes comprehensive systems like the pressurizing or cover gas system, steam generators, coolant piping, fluid drive circulation system, in-system diagnostic instrumentation, and metering, as well as main steam piping to turbine control and isolation valves and feedwater piping, highlighting the intricate and multifaceted nature of the coolant systems in maintaining reactor operation and safety. Does not include the initial reactor coolant load - this is handled in Cost Category 27 Special Materials (for all types of coolant). Schematic is shown in Fig. 10. Consists of:

- Cost Category 22.02.00.00 Main heat transfer and transport systems
- Cost Category 22.02.01.00 Primary coolant system
- Cost Category 22.02.01.01 Pumps and motor drives (modular & nonmodular)
- Cost Category 22.02.01.02 Piping
- Cost Category 22.02.01.03 Heat exchangers
- Cost Category 22 02.01.04 Tanks (dump, make-up, clean-up, trit., hot storage)
- Cost Category 22.02.01.05 Clean-up system
- Cost Category 22.02.01.06 Thermal insulation, piping & equipment
- Cost Category 22.02.01.07 Tritium extraction
- Cost Category 22.02.01.08 Pressurizer
- Cost Category 22.02.01.09 Other
- Cost Category 22.02.03.00 Secondary coolant system
- Cost Category 22.02.03.01 Pumps and motor drives (modular & non-modular)

- Cost Category 22.02.03.02 Piping
- Cost Category 22.02.03.03 Heat exchangers
- Cost Category 22.02.03.04 Tanks (dump, make-up, clean-up, trit., hot storage)
- Cost Category 22.02.03.05 Clean-up system
- Cost Category 22.02.03.06 Thermal insulation, piping & equipment
- Cost Category 22.02.03.07 Tritium extraction
- Cost Category 22.02.03.08 Pressurizer
- Cost Category 22.02.03.09 Other
- Cost Category 22.02.04.00 Thermal storage system

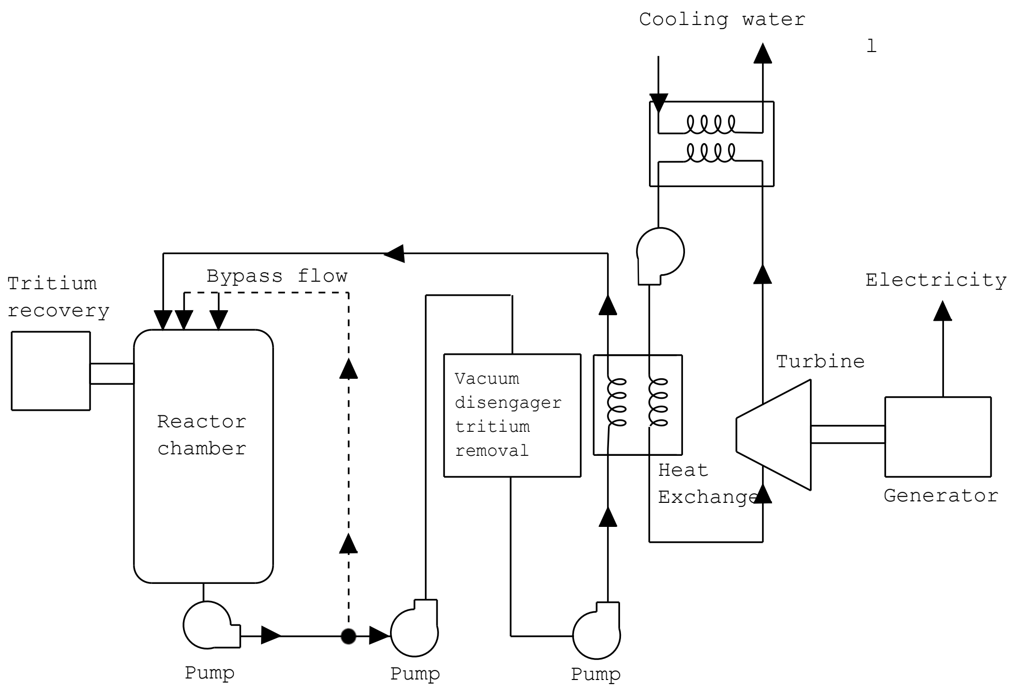

Figure 10: Main coolant system.

This concept uses Lead Lithium (PbLi) as the primary coolant, and water as the secondary coolant. The costs are , relative to thermal power, the primary coolant systems total 106 M USD, secondary total 0 M USD, resulting in a system cost of 106 M USD.

6.2.3 Cost Category 22.03: Auxiliary Cooling Systems

The Cost Category 22.03, originally designated for auxiliary cooling, was primarily considered for cryogenic cooling systems related to magnets. It has been recommended that all cooling services, including those initially covered under Account 22.03, be directly associated with the specific account requiring those services. In this instance, the auxiliary cooling services for the magnets are to be included under Cost Category 22.02.04, which deals with Cryogenics for Plasma Confinement. Auxiliary Cooling Systems therefore currently consists of:

- Cost Category 22.03.04 Power Supply and Cooling System:

- Cost Category 22.03.04.01 Refrigeration: Systems for maintaining low temperatures in critical areas.
- Cost Category 22.03.04.02 Piping: Network of pipes for the distribution of cooling fluids.
- Cost Category 22.03.04.03 Fluid Circulation Driving System: Mechanisms for propelling cooling fluids throughout the system.
- Cost Category 22.03.04.04 Tanks: Storage units for cooling fluids or refrigerants.
- Cost Category 22.03.04.05 Purification: Processes to ensure the purity and effectiveness of cooling fluids.

- Cost Category 22.03.05 Other Cooling Systems: Additional cooling systems not categorized elsewhere.

Costs are expressed as 30% of the power supply costs to give a total of $ 2.9 M.

6.2.4 Cost Category 22.04: Radioactive Waste Treatment

In the realm of managing radioactive materials in fusion energy systems, several key processes are pivotal. Firstly, the purification of the liquid breeder or heat transfer media is essential to maintain the concentration of radioactive products, specifically primaryC, within safe limits. Radioactive materials necessitate careful classification for either recycling, clearance, or disposal in appropriate off-site repositories. According to El-Guebaly et al. in "Goals, Challenges, and Successes of Managing Fusion Active Materials," [**?**] radioactive materials must be prepared for either recycling, clearance, or proper disposal in designated off-site repositories. The processing equipment is likely located in the Hot Cell building. This equipment encompasses remote handling tools, storage tanks, pumps, piping, valves, heat exchangers, heaters, condensers, gas strippers, compressors, chemical reactors, evaporators, ion exchange subsystems, filters, traps, and separators. The on-site system is designed for basic processing and management, not for extensive processing. It does not serve as the primary system for online separation of Deuterium and Tritium fuel isotopes, which is handled by the Fuel Handling and Storage system (Account 22.5). Nonetheless, cost-effectively recovered tritium, such as from detritiation processes, is reintegrated into the Fuel Handling and Storage system for further refinement.

Consists of:

- Cost Category 22.04.01 liquid waste processing and equipment. This involves purifying the liquid breeder or heat transfer medium, maintaining the concentration of radioactive products, in this case primaryC, below specific safety limits.

- Cost Category 22.04.02 gaseous wastes and off-gas processing system. It involves capturing, filtering, and treating off-gases to remove contaminants and reduce radioactive emissions. This process is vital for minimizing environmental impact and adhering to stringent safety standards.

- Cost Category 22.04.03 solid waste processing equipment. It involves the segregation, treatment, and preparation of solid wastes for safe storage or disposal. This process is essential for ensuring long-term environmental safety and compliance with regulatory requirements.

It's important to note that fusion reactors produce less long-lived radioactive waste compared to fission reactors. The IAEA notes that fusion reactor waste is primarily composed of activated structural materials, and the volume of high-level waste is significantly less [21]. This makes waste management in fusion reactors potentially more manageable and less hazardous over the long term. The development of low-activation structural materials remains a key enabling technology for fusion implementation and public acceptance.

Costs are scaled linearly with neutron power. The resultant total cost is $ 5.1 M. Referencing Table D.1. PWR-12 cost breakdown using the GIF/EMWG COA from a joint report by the OECD Nuclear Energy Agency and the International Atomic Energy Agency: 'Measuring Employment Generated by the Nuclear Power Sector' OECD 2018 NEA No. 7204 NUCLEAR ENERGY AGENCY.

Cost Category	ITER Costs (M USD)	Scaled Costs (M USD)
22.05.01	29.3	13.9
22.05.02	10.0	4.8
22.05.03	32.2	15.3
22.05.04	14.0	6.7
22.05.05	32.6	15.6
22.05.06	68.0	32.4
22.05.00	186.0	88.7

Table 7: Breakdown of fuel handling and storage subsystems for ITER and the concept presented in this report.

6.2.5 Cost Category 22.05: Fuel Handling and Storage

The fuel handling and storage system in fusion reactors is responsible for the online processing of fuel isotopes, encompassing extraction, recovery, purification, preparation, and storage. Separate from fuel injection, this system processes materials from various sources including liquid breeders, chamber gases, and tritium-bearing streams, ensuring safe tritium levels in power core and heat transfer fluids under normal and emergency conditions.

While most equipment is commercially available, heightened reliability and integrity standards, especially for tritium-containing materials, lead to higher costs. For example, atmospheric tritium recovery systems are costly due to the need for rapid and efficient tritium concentration reduction. This account's functionality and requirements are well-understood from existing and developing fusion facilities, with revisions from previous versions. Cost increases are expected due to higher reliability demands for power plant applications, but this may be offset by learning curve effects in subsequent plant constructions. The recommended Life-Cycle Analysis (LSA) factors for all accounts are 0.85 (LSA1) and 0.94 (LSA 2), reflecting these considerations.

Consists of:

- Cost Category 22.05.01 Chamber exhaust gas handling and processing equipment (H&P). Equipment for filtering, treating, and safely managing exhaust gases from the reactor chamber.
- Cost Category 22.05.02 Purge and cover gas H&P. Systems used for handling and processing gases that protect fuel and reactor components from contamination.
- Cost Category 22.05.03 Primary coolant stream H&P. Equipment involved in the handling and processing of the primary coolant stream, essential for maintaining coolant integrity.
- Cost Category 22.05.04 Purification and isotope separation. Systems for the purification of reactor materials and the separation of isotopes, vital for fuel quality and waste management.
- Cost Category 22.05.05 Tritium, deuterium and DT storage. Facilities for storing tritium, deuterium, and DT mixtures, which are integral to nuclear fusion reactions.
- Cost Category 22.05.06 Atmospheric tritium recovery. Systems designed to recover tritium from the atmosphere within the reactor facility, crucial for environmental protection and resource conservation

Various previous systems and cost analyses are available as extrapolation points, in particular For this report, ITER is used as the benchmark, then linearly scaled by P_E, scaled for inflation, and a learning curve credit of 0.8 is applied, equating to a factor of 0.5 for a nth-of-a-kind system. The resulting total cost is 88.7 M USD. A breakdown of costs for each susbsystem for ITER and this concept can be found in table 7.

6.2.6 Cost Category 22.06: Other Heat Island Equipment

This Cost Category encompasses various essential components and systems of a reactor plant that are not specifically categorized elsewhere. This category is crucial for the comprehensive coverage of all necessary equipment in reactor plant operations and maintenance, safety and compliance. This category includes, but is not limited to:

- Cost Category 22.06.01 Maintenance Equipment
- Cost Category 22.06.01.01 Blanket Maintenance Equipment. Tools and systems for maintaining the blanket.
- Cost Category 22.06.01.02 Components Rotated into Service. Spare components used to replace operational ones during maintenance.
- Cost Category 22.06.01.03 Other. Miscellaneous equipment required for general maintenance activities.
- Cost Category 22.06.02 Special Heating Systems
- Cost Category 22.06.02.01 Start-up Systems. Equipment used for heating processes during start-up phases.
- Cost Category 22.06.03 Coolant Receiving, Storage and Make-up Systems.
- Cost Category 22.06.03.01 Coolant Handling. Systems for receiving, storing, and preparing coolant.
- Cost Category 22.06.04 Gas Systems
- Cost Category 22.06.04.01 Gas Handling and Management. Systems for managing gases used in or produced in the Heat Island.
- Cost Category 22.06.05 Building Vacuum Systems.
- Cost Category 22.06.05.01 Vacuum Maintenance. Systems to maintain vacuum conditions within certain Heat Island areas, crucial for operational integrity and safety.

Due to the broad nature of this cost category, bottom-up calculations are difficult to perform, without precise details and engineering designs. For costing purposes, we allocate 10% of the cost of the Heat Island component (cost category 22.1) costs to this category, and the total budget is 8 M USD.

6.2.7 Cost Category 22.07: Instrumentation and Control

For a NOAK plant, it is anticipated that the I&C technologies will be much more advanced than current capabilities. This system will include the power core instrumentation and control, radiation and monitoring equipment, isolated indicating and recording equipment, data acquisition and recording, communication equipment, benchboards, panels, racks, process computers, I&C tubing and fittings, instrumentation, and software, consisting of the following cost items:

- Cost Category 22.07.01 Heat Island Instrumentation and Control Equipment. Instrumentation and control equipment to monitor and manage the plasma and all its related functional equipment contained in the Heat Island (includes all equipment in Account 22).
- Cost Category 22.07.02 Radiation Monitoring Equipment. Monitoring equipment to ensure safe levels of radiation are maintained during routine plant operations, determining the tritium inventories in all plant components, and analyze/report any fault or accidental release of radioactive materials.
- Cost Category 22.07.03 Steam Turbine Control Equipment
- Cost Category 22.07.04 Other Major Component Control Equipment
- Cost Category 22.07.05 Signal Processing Equipment
- Cost Category 22.07.06 Control Boards, Panels, & Racks
- Cost Category 22.07.07 Distributed Control System Equipment

- Cost Category 22.07.08 Instrument Wiring and Tubing
- Cost Category 22.07.09 Other Instrumentation & Controls Equipment

We use the NETL Case B31A Category 12 Instrumentation and Control for a 727MW Net power plant with bare erected costs of $14M representing 19$/kW. Instrumentation in the heat island remains a major cost driver for FOAK, but by NOAK should come pre-packaged as multiple instrument packages, totalling $ 50M, giving 88$/kW. We calculate $ 21 M for this cost category.

6.3 Cost Category 23: Turbine Plant Equipment

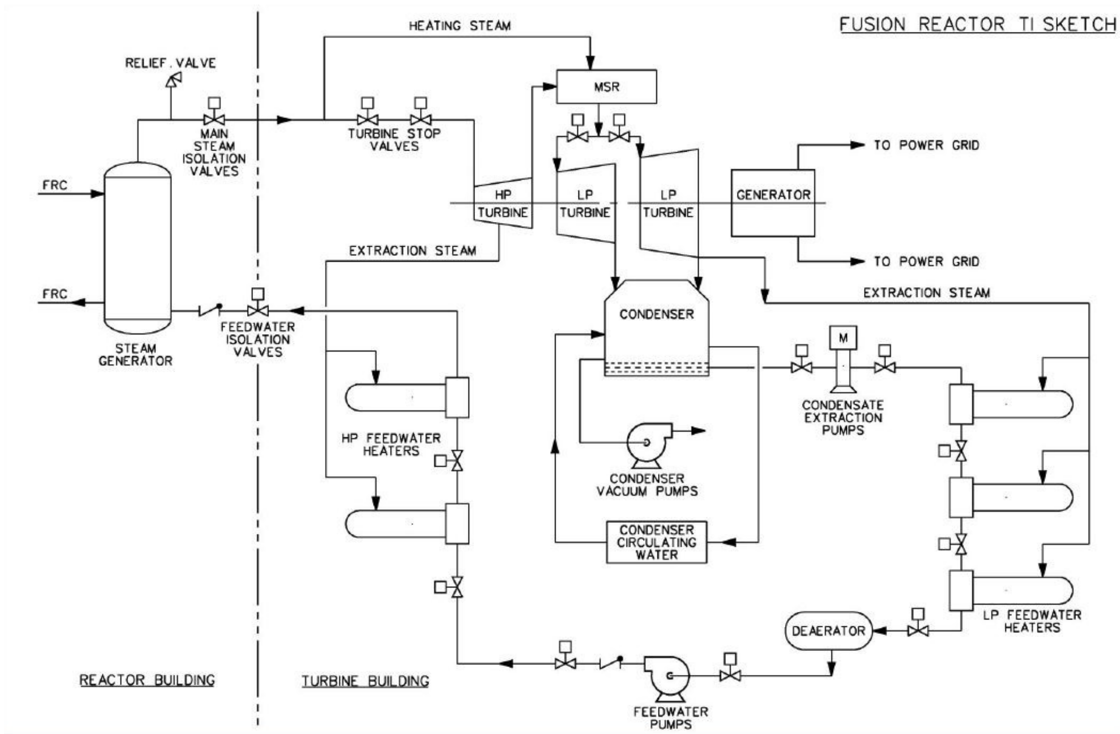

Figure 11: Turbine island, per the 2017 ARPA-E study [3].

Turbine Generator Equipment: This category assumes that electricity is the primary product. The categories below apply mostly to a steam-driven turbine; however, similar categories would exist for gas-driven turbines. This account includes all process equipment and systems associated with the plant output.

- Cost Category 23.10.00 Turbine Generator(s): Includes turbine generator plus associated mountings, main steam control and isolation valves, lubrication system, gas systems, moisture separator, and drain system, excitation system, and controls. Main steam piping is in Account 22.20.00.
- Cost Category 23.30.00 Condensing Systems: Includes condenser equipment, the condensate system, the gas removal system, the turbine bypass system, condenser-cleaning system, and piping from condenser to the feedwater heating system (Cost Category 23.40.00). Condensate polishing is in Account 23.50.00
- Cost Category 23.30.00 Feed Heating Systems: Includes the feed heating system, feedwater heaters, feedwater system piping, the extraction steam system, and the feedwater heater vent and drain system. Piping to the steam generator continues with Cost Category 22.20.00.

- Cost Category 23.50.00 Other Turbine Plant Equipment: Includes piping system, turbine auxiliaries, closed cooling water system, demineralized water make-up system, chemical treatment system, and neutralization system. The cooling towers are in Cost Category 26.00.00.

- Cost Category 23.60.00 Instrumentation and Control (I&C): Includes turbine generator control equipment, process computer, and BOP I&C, including software, tubing, and fittings. Cables are in Cost Category 24.6.

- Cost Category 23.70.00 Turbine Plant Miscellaneous Items: Includes painting, welder qualification, and turbine plant insulation.

Cost basis is NETL baseline Case B12A, Cost Category 8: Steam Turbine and Accessories Bare erected cost excluding engineering, construction management, and contingency $150 million in source table for Coal Plant / 686 MW electric capacity in source table = $219/kW applied to gross thermal power, to give a total of $ 182 M.

6.4 Cost Category 24: Electric Plant Equipment

Consists of the switchgear, station service eqiupment, switchboards, protective equipment, electrical structures and wiring containers, power and control wiring and electrical lighting. Accounts 21 through 23 all have interfaces with the power plant electrical service system and its associated equipment. This equipment is located both inside and outside the main reactor/BOP buildings.

- Cost Category 24.01.00 Switchgear: Includes switchgear for the generator and station service, generator transformer, auxiliary transformer, and connecting bus (typically to 10kV).

- Cost Category 24.02.00 Station Service Equipment: Includes substations, auxiliary power sources, load centers, motor control centers, and station service and startup equipment, transformers, and bus ducts (typically to 500V).

- Cost Category 24.03.00 Switchboards: Includes control panels, auxiliary power and signal boards, batteries, DC equipment, and non-uninterruptible power.

- Cost Category 24.04.00 Protective Systems Equipment: Includes grounding systems, lightning protection, cathodic protection, heat tracing, freeze protection equipment, radiation monitoring, environmental monitoring equipment, raceway, cable, and connections. Excludes communication systems and building services such as lighting, HVAC, and fire protection (which are included in their respective building accounts).

- Cost Category 24.05.00 Electrical Raceway Systems: Includes cable tray, exposed conduits, embedded conduits, underground conduit and duct systems, and the fittings, supports, covers, boxes, access holes, ducts, and accessories for the scheduled cable systems. Excludes raceways for protective systems (Account 24.4) and building services electrical systems (which are included in their respective building accounts).

- Cost Category 24.06.00 Power and Control Cables and Wiring: Typically includes all scheduled cable systems and their associated fittings such as cable, straps, attachments, terminations, wire lugs, cable numbers, and wire numbers. Excludes lighting, communication, and other protection systems.

Cost basis is the NETL baseline Case B12A, Cost Category 11: Accessory Electric Plant Bare erected cost excluding engineering, construction management, and contingency $37 million in source table for Coal Plant / 686 MW electric capacity in source table = $54/kW applied to gross thermal power, to give a total of $ 45 M.

6.5 Cost Category 25: Miscellaneous Plant Equipment

Consists of items not in the other cost categories.

- Cost Category 25.01.00 Transportation and Lift Equipment: Includes cranes and hoists. (Elevators are in their respective building accounts.)
- Cost Category 25.02.00 Air, Water, Plant Fuel Oil, and Steam Service Systems
- Cost Category 25.03.00 Communications Equipment: Includes telephones, radio, CCTV, strobe, public address, enunciator, and electronic access control and security systems.
- Cost Category 25.04.00 Furnishing and Fixtures: Includes safety equipment, chemical laboratory equipment, instrument shop equipment, maintenance shop equipment, office equipment and furnishings, change room, and dining facility equipment.

Costs are scaled relative to thermal power, with factor 0.038$/MW, to give a total of $ 32 M.

6.6 Cost Category 26: Heat Rejection

This cost scales primarily with the difference between the thermal power and the gross electric power (heat rejection). Scales per Delene as overall system with spares allowance. Subsystems include water intake, circulating water systems, cooling towers, and other systems which reject heat to the atmosphere. The cooling towers provide the ultimate heat sink for the condenser cooling and service water systems. For this cost study, the cooling towers are assumed to be mechanical draft (fans) rectangular cell-type towers. On-site power (standby power) will be available to sustain operation of the plant following a loss of off-site power. The site is assumed to be located near a surface water source for makeup water needs of the plant. The power plant is assumed to be located in a warm/humid climate at a nominal plant elevation of 100 feet above sea level.

- Cost Category 26.01.00 Structures: Includes structures for the make-up water and intake, the circulating water-pump house, the make-up water pre-treatment building, and the cooling towers.
- Cost Category 26.02.00 Mechanical Equipment: Includes the heat rejection mechanical equipment such as circulating water pumps, piping, valves, mechanical draft cooling towers, water treatment plant, intake water pumps, screens, and filters. Excludes condensers and natural draft towers.

Cost basis: NETL baseline NGCC costs. Exhibit 5-7, Cost Category 9: Cooling System. Bare erected cost excluding engineering, construction mgmt, and contingency $28 million in source table for NGCC / 263 MW electric capacity in source table for NGCC = $107/kW to give a total of 78 M USD.

6.7 Cost Category 27: Special Materials

This account contains the costs of specific types of materials that are introduced into the fusion power plant immediately before the initiation of testing and validation procedures. These materials are characterized within conventional categories, encompassing coolants, cover gases designed for material handling systems and architectural components, breathable air containing specific compositions, as well as gases or liquids that have not been pre-loaded into the existing plant systems. It is essential to classify these materials as intrinsic components of the capital equipment, distinct from the diverse plant systems, and therefore not subject to acquisition via construction loans. The replenishment process for these materials is to be delineated as an operational expense. Among the specialized materials encompassed are eutectic salts, treated water, liquid metals, cryogenic liquids, gases with unique properties, and potentially select solid materials, moderator and/or reflector materials that are non-fuel and non-structural in nature. Special heat transfer fluids, which are different from natural water, and may be in the form of gases or liquids. This includes primary and secondary coolant, intermediate loop heat transport fluid, and turbine cycle working fluids. The initial supply of oil, lubricants, ion exchange resins, boric acid, and gases such as N_2, O_2, He, and CO_2. This category is further broken down into sub-categories:

Purity	Cost factor
90 %	1
99 %	2
99.9 %	4
99.99 %	8
99.999 %	16

Table 8: Purity cost factors.

- Cost Category 27.01.00 Primary Coolant. Costs are based on volumes. For a primary coolant of Lead Lithium (PbLi) with a volume of 25 cubic meters, a cost of $ 2 M is obtained.
- Cost Category 27.02.00 Secondary Coolant. Costs are based on volumes. For a primary coolant of secondaryC with a volume of secondaryV cubic meters, a cost of $ 0 M is obtained.
- Cost Category 27.03.00 Not used
- Cost Category 27.04.00 Not used
- Cost Category 27.05.00 Turbine cycle working fluid.
- Cost Category 27.06.00 Initial materials, including oil, lubricants, resins for ion exchanger, and boric acid.

Note that for natural lithium, there is a cost per tonne quoted in the materials look-up table, Table 3. If the lithium needs to be enriched or purified, the costs do not scale linearly with purity. We include a scaling with the purity therefore as Table 8 suggests.

Total costs for special materials is the summation over all the sub categories, giving $ 3 M.

6.8 Cost Category 28: Digital twin / Simulator

Simulator(s): This account includes the development of new digital twins or simulators. Digital twins play a pivotal role in enhancing the safety, efficiency, and overall operational effectiveness of nuclear power plants. By creating real-time, high-fidelity digital replicas of these complex facilities, operators can continuously monitor and simulate plant behavior, predict potential issues, and optimize maintenance schedules. This technology enables proactive decision-making, reduces downtime, and ensures compliance with stringent safety standards, ultimately contributing to the reliable and sustainable operation of nuclear power plants.

Cost basis is the estimated annual license fee of 5 M USD for a comprehensive digital twin platform for the fusion power plant, based on quotes from nTtau Digital LTD.

6.9 Cost Category 29: Contingency on Direct Capital Costs

This account includes the assessment of additional cost that might be necessary to achieve the desired confidence level for the direct costs not to be exceeded. Contingency is usually applied at an aggregated level, although its determination may include applying contingencies to individual high-cost-impact items in the estimate. Contingency factors range from 1 to 0.05 depending on the maturity of the subsystem, so for a FOAK Fusion Pilot Plant, the factor may be as high as 1 for all of the major subsystems included in the fusion power core, but closer to 0.05 for systems in the Balance of Plant.

Here we apply a factor of 0.1 to all of the systems in Cost Category 20, giving a total of 0 M. However for NOAK, we omit this cost.

7 Cost Category 30: Capitalized Indirect Service Costs (CISC)

Cost Category 31 – Field Indirect Costs

This Cost Category includes cost of construction equipment rental or purchase, temporary buildings, shops, laydown areas, parking areas, tools, supplies, consumables, utilities, temporary construction, warehousing, and other support services. Cost Category 31 also includes:

- Temporary construction facilities, such as site offices, warehouses, shops, trailers, portable offices, portable restroom facilities, temporary worker housing, and tents.
- Tools and heavy equipment used by craft workers and rented equipment such as cranes, bulldozers, graders, and welders. Typically, equipment with values of less than $1,000 are categorized as tools.
- Transport vehicles rented or allocated to the project, such as fuel trucks, flatbed trucks, large trucks, cement mixers, tanker trucks, official automobiles, buses, vans, and light trucks.
- Expendable supplies, consumables, and safety equipment.
- Cost of utilities, office furnishings, office equipment, office supplies, radio communications, mail service, phone service, and construction insurance.
- Construction support services, temporary installations, warehousing, material handling, site cleanup, water delivery, road and parking area maintenance, weather protection and repairs, snow clearing, and maintenance of tools and equipment.

Previously calculated as 6 percent of total direct costs based on Field Office Engineering and Services Table 3.2-VII of [22]. Min $130/kW, Average $158/kW, Max $192/kW, to give a total of $ 92 M for a construction time of 0.5 years. Reduction strategies: Same as strategies listed above for construction services and materials (91) and home office engineering and services (92).

Bottom up example: 10 engineers and managers on average during construction period, with 2000 hours of work annually per engineer and manager = 60,000 man-hours of work $150/hour average rate = $3 million, with 150 MW for plant capacity = $0.02/W/annum, giving total cost of $ 3 M for a construction time of 0.5 years.

Cost Category 32 – Construction Supervision

This Cost Category covers the direct supervision of construction (craft-performed) activities by the construction contractors or direct-hire craft labor by the A/E contractor. The costs of the craft laborers themselves are covered in the labor-hours component of the direct cost in Cost Categorys 21 through 28 or in Cost Category 31. This Cost Category covers work done at the site in what are usually temporary or rented facilities. It includes non-manual supervisory staff, such as field engineers and superintendents. Other non-manual field staff are included with Cost Category 38, PM/CM Services Onsite.

This cost category supersedes the Pre 2007 indirect cost categories 91 Construction Service and Materials

Previous values: 12% of total direct costs based on Schulte et al. (1978) and LSA of 2, giving a total of 3 M USD. Min $261/kW, Average $316/kW, Max $384/kW.

Here we estimate from the bottom up: 25 managers on average during construction period. 2000 hours of work annually per engineer and manager = 50,000 man-hours of work per annum $150/hour average rate=$7.5 million for a 150 MW plant capacity = $0.05/W/annum, to give a total of 8 M USD for a construction time of 0.5 years.

Cost Category 33 – Commissioning and Start-up Costs

This Cost Category includes costs incurred by the A/E, reactor vendor, other equipment vendors, and owner or owner's representative for startup of the plant including:

- Startup procedure development
- Trial test run services (Cost Category 37 in the IAEA Account system)
- Commissioning materials, consumables, tools, and equipment (Cost Category 39 in the IAEA Account system)

The utility's (owner's) pre-commissioning costs are covered elsewhere in the TCIC sum as a capitalized owner's cost (Cost Category 40).

Cost Category 34 – Demonstration Test Run

This Cost Category includes all services necessary to operate the plant to demonstrate plant performance values and durations, including operations labor, consumables, spares, and supplies.

Cost Category 35 – Design Services Offsite

This Cost Category covers engineering, design, and layout work conducted at the A/E home office and the equipment/reactor vendor's home office. Often pre-construction design is included here. These guidelines use the IAEA format for a standard plant (and equipment) design/construction/startup only and not the FOAK design and certification effort. (FOAK work is in the one-time deployment phase of the project and not included in the standard plant direct costs.) Design of the initial full size (FOAK) reactor, which will encompass multiple designs at several levels (pre-conceptual, conceptual, preliminary, etc.), will be a category of its own under FOAK cost. This Cost Category also includes site-related engineering and engineering effort (project engineering) required during construction of particular systems, which recur for all plants, and quality assurance costs related to design.

Previously this was Cost Category 92 - Home Office Engineering Services. Home Office Engineering and Services Table 3.2-VII of [22].

Adaptations to generic plant design for the specific construction project by off-site engineers, and other necessary off-site engineering support. Note: In this standard framework for nuclear plant cost accounting, only activities for specific construction projects at particular sites should be included in this code or any other. Initial work by plant vendors to design their generic concepts and license them with the NRC should be excluded from project-specific cost accounting. Plant vendors can recover their initial design and licensing costs over several projects (e.g., first through fifth) by applying an adder, perhaps as part of overhead for profitability, to the project-specific costs.

5% of total direct costs based on Schulte et al. (1978). Min \$113/kW, Average \$137/kW, Max \$166/kW, to give a total of 80 M USD. Reduction strategy includes standardized and reusable plant design; avoidance of regulatory or plant owner modifications to generic design.

Example basis: 15 engineers and managers on average during construction period. 1000 hours of work annually per engineer and manager = 30,000 man-hours of work per annum at \$150/hour average rate = \$4.5 million per annum with 150 MW for plant capacity = \$0.03/W/annum and a total of 5 M for a construction time of 0.5 years.

Cost Category 36 – PM/CM Services Offsite

This Cost Category covers the costs for project management and construction management support on the above activities (Cost Category 31) taking place at the reactor vendor, equipment supplier, and A/E home offices.

Cost Category 37 – Design Services Onsite

This Cost Category includes the same items as in Cost Category 35, except that they are conducted at the plant site office or onsite temporary facilities instead of at an offsite office. This Cost Category also includes additional services such as purchasing and clerical services.

Cost Category 38 – PM/CM Services Onsite

This Cost Category covers the costs for project management and construction management support on the above activities taking place at the plant site. It includes staff for quality assurance, office administration, procurement, contract administration, human resources, labor relations, project control, and medical and safety-related activities. Costs for craft supervisory personnel are included in Cost Category 32.

Cost Category 39 – Contingency on Support Services

This Cost Category includes an assessment of additional cost necessary to achieve the desired confidence level for the support service costs not to be exceeded.

8 Cost Category 40: Capitalized Owner's Cost (COC)

Previously was Cost Category 94 and was calculated as 23 % of total direct costs based on Owner's Cost - Table 3.2-VII of [22] and LSA of 2. Costs of $ 185 M are estimated. Reduction strategies include streamlined processes for staff recruitment etc.

Cost Category 41 – Staff Recruitment and Training

This Cost Category includes costs to recruit and train plant operators before plant startup or commissioning activities (Cost Category 33), or demonstration tests (Cost Category 34).

Cost Category 42 – Staff Housing

This Cost Category includes relocation costs, camps, or permanent housing provided to permanent plant operations and maintenance staff.

Cost Category 43 – Staff Salary-Related Costs

This Cost Category includes taxes, insurance, fringes, benefits, and any other salary-related costs.

Cost Category 44 – Other Owner's Costs

Cost Category 49 – Contingency on Owner's Costs

This Cost Category includes an assessment of additional costs necessary to achieve the desired confidence level for the capitalized owner's costs not to be exceeded.

9 Cost Category 50: Capitalized Supplementary Costs (CSC)

Cost Category 51 – Shipping and Transportation Costs

This Cost Category includes shipping and transportation costs for major equipment or bulk shipments with freight forwarding. Budget is $ 8 M.

Cost Category 52 – Spare Parts

This Cost Category includes spare parts furnished by system suppliers for the first year of commercial operation. It excludes spare parts required for plant commissioning, startup, or the demonstration run. Budget is $ 34 M, which is calculated on the basis of 10 % of the direct cost components in the BOP (note that Scheduled Replacement Costs for the heat island are calculated in Cost Category 75.

Cost Category 53 – Taxes

This Cost Category includes taxes associated with the permanent plant, such as property tax, to be capitalized with the plant. Budget is $ 100 M.

Cost Category 54 – Insurance

This Cost Category includes insurance costs associated with the permanent plant to be capitalized with the plant. Budget is $ 1 M.

Cost Category 55 – Initial Fuel Load

This Cost Category covers fuel purchased by the utility before commissioning, which is assumed to be part of the TCIC. For DT systems this will include the initial tritium startup costs. Due to its scarcity, tritium costs $109,570 to $169,570 per gram in 2016 USD. Therefore, the onetime start-up tritium cost ranges from $22 M to $34 M (2016 USD) for a standard 150 MWe FPP.

We allocate a budget of $ 0 M for the initial fuel load.

Cost Category 58 – Decommissioning Costs

This Cost Category includes the cost to decommission, decontaminate, and dismantle the plant at the end of commercial operation, if it is capitalized with the plant. This cost used to be broken out as a separate line item in the cost of electricity calculation.

The D&D allowance are assumed flat and constant for the systems under consideration here, and we base the amount on the following:

```
Decontamination and Decommissioning Allowance
TFTR (51 MWth) $36.7M (Actual 2002$)
Generic Magnetic Fusion Reactor 0.5 M/kWh (1983$)
Light Water Reactor $307M - $819M (Actual 2013$)
Generic Magnetic Fusion Reactor Revisited, J Sheffield/S Milora,
ANS Fusion Science and Technology Vol.70, 14-35, July 2016.
Decommissioning of the Tokamak Fusion Test Reactor,
Perry/Chrzanowski/Gentile/Parsells/Rule/Strykowsky/Viola,
Princeton Plasma Physics Laboratory, PPPL-3895, October 2003.
Costs of Decommissioning Nuclear Power Plants, NEA No. 7201. OECD 2016.
```

Prior to 2007, these costs were included in the LCOE as a budget of 0.5mills/kWh. Here, we provide a budget estimate for the decommissioning costs of $ 70 M, which is consistent with current D&D costs for SMRs.

Cost Category 59 – Contingency on Supplementary Costs

This Cost Category includes an assessment of additional cost necessary to achieve a desired confidence level for the capitalized supplementary costs not to be exceeded. The contingency for the initial core load should not be applied to this item, because the contingency is already embedded in the fuel cycle costs from the fuel cycle model.

Budget is $ 0 M, based on 10 % of the sum of the CSC.

10 Cost Category 60: Capitalized Financial Costs (CFC)

Cost Category 61 – Escalation

This Cost Category is typically excluded for a fixed year, constant dollar cost estimate, although it could be included in a business plan, a financing proposal, or regulatory-related documents. Formerly Cost Category 98 pre 2007.

Escalation during Construction (EDC), which results from increases (or decreases) in the costs of labour or materials due to inflation (or recession and deflation). Table 3.2-X of [22], to give a total of $ 148 M.

Cost Category 62 – Fees

This Cost Category includes any fees or royalties that are to be capitalized with the plant.

Fees or royalties are not included.

Cost Category 63 – Interest During Construction (IDC)

This Cost Category is discussed in Chapter 7 of the G4EMWG guidelines. IDC is applied to the sum of all up-front costs (i.e., Cost Category 10 through 50 base costs), including respective contingencies. These costs are incurred before commercial operation and are assumed to be financed by a construction loan. The IDC represents the cost of the construction loan (e.g., its interest). Formerly Cost Category 97 (pre 2007).

Compound interest on debt and rate of return on equity investments during construction period depends on weighted average cost of capital between interest rate and equity return as well as construction duration for compounding calculation. Previously, 49 percent of total direct costs based on Interest during Construction (IDC) Table 3.2-X of [22]. Total based on LSA of 2 was $ 421 M. Min $1,075/kW, Average $1,302/kW, Max $1,582/kW. Reduction strategies can include minimizing interest rate and equity return: Government financing or loan guarantees; minimal project risk to assure lenders and equity investors. Minimize construction duration: Streamlined planning and processes; completed designs before construction starts and no subsequent changes; minimal construction project risks to avoid rework and delays.

Calculated here: 7 percent weighted average cost of capital = 10 percent added for interest during construction (approximate calculation - accounts for spreading construction expenditures across construction time) $99/kW. Total cost here is 32 M USD.

Cost Category 69 – Contingency on Capitalized Financial Costs

This Cost Category includes an assessment of additional cost necessary to achieve the desired confidence level for capitalized financial costs not to be exceeded, including schedule uncertainties.

Currently not calculated.

11 Cost Accounting Structure

	Cost (M USD)	% of TCC	MARS Cost (M USD)	MARS % of TCC
10. Pre-construction	42	2	0.0	0.0
20. Direct Costs	1538	71	7100.0	79.0
21. Structures/Site	258	12	1200.0	13.0
22. Reactor Plant Equip.	672	31	2700.0	29.0
22.1 Reactor Equip.	441	20	-	-
22.1.1 First Wall & Blanket	150	7	-	-
22.1.2 Shield	80	4	-	-
22.1.3 Magnets	0	0	2200.0	24.0
22.1.4 Compression (Jets)	25	1	-	-
22.1.5 Primary Structure	0	0	-	-
22.1.6 Vacuum System	20	1	-	-
22.1.7 Power Supplies	111	5	-	-
22.1.8 Target Plasma	10	0	-	-
22.1.9 Direct E. Conv.	0	0	-	-
22.1.11 Assembly and installation	5	0	-	-
22.2 Main Heat Transfer	106	5	-	-
22.3 Auxiliary Cooling	3	0	-	-
22.4 Rad. Waste Treat.	5	0	140.0	1.5
22.5 Fuel Processing	89	4	140.0	1.5
22.6 Other plant equipment	8	0	-	-
22.7 Instrumentation and control	21	1	160.0	1.8
23. Turbine Plant Equip.	182	8	790.0	8.7
24. Electric Plant Equip.	45	2	-	-
25. Misc. Plant Equip.	32	1	-	-
26. Heat Rejection	78	4	-	-
27. Special Materials	3	0	-	-
28. Digital Twin	5	0	-	-
29. Contingency	0	0	-	-
30. Capitalized Indirect Service Costs (CISC)	16	1	51.0	0.56
40. Capitalized Owner's Cost (COC)	185	8	610.0	6.7
50. Capitalized Supplementary Costs (CSC)	213	10	700.0	7.7
60. Capitalized Financial Costs (CFC)	180	8	590.0	6.5
Total Capital Cost (TCC):	2173	100	9100.0	100.0

Table 9: Cost accounts with updated costs and codes.

12 Cost of Electricity

Contributors to LCOE projections are given by:

$$LCOE[\$MWh] = \frac{(C_{AC} + (C_{OM} + C_F)(1+y)^Y)}{(8760.P_E.p_a)} \tag{6}$$

where C_{AC} [USD/year] is the annual capital cost charge (entailing the total capital cost of the plant (TCC (USD) multiplied by the Fixed Charge Rate (FCR (/year)), C_{OM} [USD/year] is the annual operations and maintenance cost, C_{SCR} [USD/year] is the annual scheduled component replacement costs, C_F [USD/year] is the annual fuel costs, y is the annual fractional increase in fuel costs over the expected lifetime of the plant Y [years], P_E [MWe] is the net electric power of the plant, p_a is the plant availability (typically 0.6-0.9). A small charge used to be imposed to build up a fund to cover end-of-life Decontamination and Decommissioning (DD), f_{DD} (USD/kWh), but the DD costs are now included in the direct capitalized costs (Cost Category 58), so are omitted here.

12.1 Cost Category 70: Annualized O&M Cost (AOC)

The Operations and Maintenance (O&M) costs was previously estimated to be 2 % of the direct cost, assessed annually. The O&M cost depends on the system complexity and the requirements of regulations, security and maintenance. The costs are computed based on look-up tables, such as the one shown in Fig. 12 at 60 USD per kilowatt-year.

```
C_OM = 60 * PE * 1000 = 38
```

Annualized O&M costs are $ 38 M.

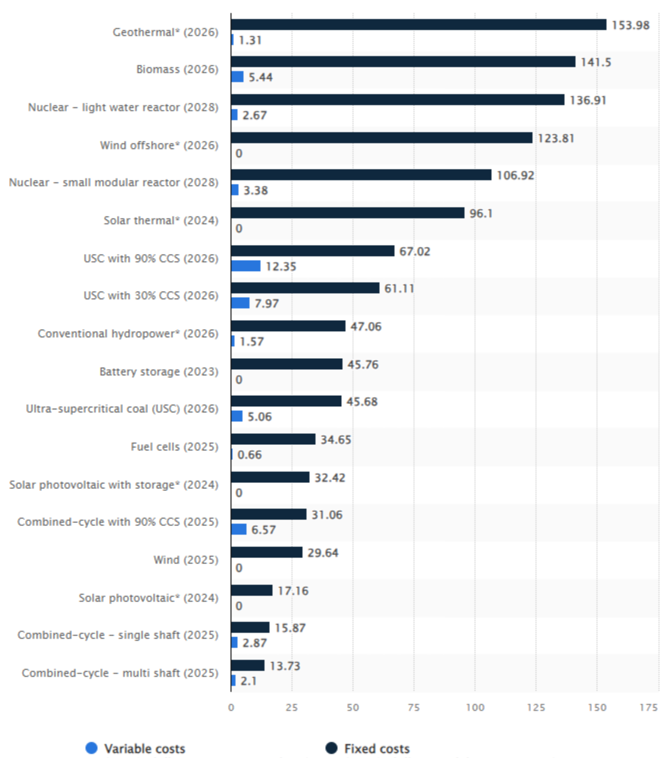

Figure 12: Operations and Maintenance costs for various kinds of power plants.

Cost Category 71 – O&M Staff

Component in the financial structuring of operational expenditures in various industries, especially those involving significant infrastructure and machinery, such as power generation, manufacturing, and utilities. The costs under Cost Category 71 predominantly include:

- **Salaries:** The regular wages paid to the O&M staff.
- **Training and Development:** Costs associated with professional development, training programs, and certifications to ensure staff is up-to-date with the latest operational and maintenance practices.
- **Overtime and Shift Allowances:** Compensation for extended work hours and shift differentials, especially in 24/7 operational setups.

Cost Category 71 is crucial in operational budgeting due to:

1. **Staffing Optimization:** The need for an optimal number of skilled staff to ensure efficient and safe operations.
2. **Direct Impact on Operational Efficiency:** The performance and availability of O&M staff directly affect operational uptime and efficiency.
3. **Regulatory Compliance:** Compliance with labor laws and industry-specific regulations regarding staffing and worker safety.

Cost Category 71 – O&M Staff encompasses all aspects of costs related to the personnel responsible for the operation and maintenance of facilities. Efficient management of this category is vital for ensuring operational excellence, staff well-being, and overall financial health of the organization.

Cost Category 72 – Management Staff

Crucial aspect of operational expenses in various sectors, especially in industries where effective management plays a pivotal role in ensuring efficient operations and achieving organizational objectives. Cost Category 72 typically encompasses:

- **Salaries:** The basic remuneration paid to operations management staff.
- **Bonuses and Incentives:** Performance-related bonuses and incentives that align management objectives with corporate goals.
- **Professional Development:** Expenses related to the continuous learning and development of management skills, including workshops, seminars, and courses.
- **Leadership Training:** Specific training focused on enhancing leadership qualities necessary for effective management.

The inclusion of Cost Category 72 in operational expenditure is significant due to:

1. **Strategic Decision Making:** Management staff play a crucial role in strategic decision-making and operational planning.
2. **Operational Efficiency:** Effective management is key to ensuring operational efficiency and organizational success.
3. **Staff Leadership:** Management staff are responsible for leading, guiding, and motivating the workforce, directly impacting productivity and morale.

Cost Category 73 – Salary-Related Costs

This Cost Category includes taxes, insurance, fringes, benefits, and any other annual salary-related costs.

Cost Category 74 – Operations Chemicals, and Lubricants

Cost Category 75 – Spare Parts

Critical element in the operational budgeting of various industries, particularly in sectors that rely on continuous and efficient machinery and equipment operation. This cost category encompasses:

- **Operational Spare Parts:** Expenses associated with spare parts used in regular operations. These do not include major equipment or capital plant upgrades.
- **Exclusion of Capitalized Items:** It specifically excludes items that would be capitalized or amortized over a period or quantity of product.

Cost Category 75 plays a significant role in Operations and Maintenance (O&M) budgeting, as it:

- **Ensures Operational Continuity:** Regular replacement and availability of spare parts are crucial for uninterrupted operations.
- **Impacts Maintenance Scheduling:** Influences the planning and scheduling of maintenance activities.
- **Affects Operational Efficiency:** Directly impacts the overall efficiency and lifespan of the operational equipment.

Cost Category 75 – Spare Parts is essential for the smooth operation and maintenance of machinery and equipment in various industries. It encompasses the costs of spare parts necessary for regular operation and scheduled component replacements, playing a vital role in operational budgeting and efficiency. Cost of any operational spare parts, excluding capital plant upgrades or major equipment that will be capitalized or amortized over some period or quantity of product.

Cost Category 76 – Utilities, Supplies, and Consumables

Cost of water, gas, electricity, tools, machinery, maintenance equipment, office supplies and similar items purchased annually.

Cost Category 77 – Capital Plant Upgrades

Upgrades to maintain or improve plant capacity, meet future regulatory requirements or plant life extensions.

Cost Category 78 – Taxes and Insurance

Property taxes and insurance costs, excluding salary related.

Cost Category 79 – Contingency on Annualized O&M Costs

This Cost Category includes an assessment of additional cost necessary to achieve the desired confidence level for the annualized O&M costs not to be exceeded.

12.2 Cost Category 80: Annualized Fuel Cost (AFC)

The fuel cost, C_F, is calculated as follows. The unit cost of deuterium as D2 is 3,700 $/kg; deuterium contributes negligibly to the COE of a fusion power plant. In the long run, the power plant is self-sufficient in terms of tritium fuel production because of the breeding capability of the blanket so that no specific tritium-fuel charge is reported. It should be recognized, however, that there is a significant cost for tritium in the direct cost of the D-T fueled fusion reactor, represented in Cost Category 22.05 Fuel Handling and Storage. Cost of the primaryC and secondaryC coolant is included in Cost Category 27 Special Materials.

Consists of:

```
m_D = 3.342*10^(-27) # (kg)
u_D = 2175 #Where u_D ($/kg) = 2175 ($/kg)
C_F = N_mod * P_NRL * 1e6 * 3600 * 8760 * u_D * m_D * p_a / (17.58 * 1.6021e-13)
```

Total annual fuel costs are $ 1 M per year.

Cost Category 81 – Refueling Operations

This Cost Category includes incremental costs associated with refueling operations.

Cost Category 84 – Fuel

This Cost Category includes annualized costs associated with the fuel cycle.

Cost Category 86 – Processing Charges

This Cost Category includes storage and processing if fuel is brought in from offsite.

Cost Category 87 – Special Nuclear Materials

This Cost Category covers materials such as heavy water, sodium, lead, helium, or other energy transfer mediums that are required on an annual basis. It includes costs associated with disposal or treatment if necessary.

Cost Category 89 – Contingency on Annualized Fuel Costs

This Cost Category includes an assessment of additional cost necessary to achieve the desired confidence level for the annualized fuel costs not to be exceeded.

12.3 Cost Category 90: Annualized Financial Costs (AFC)

Consists of: Capital recovery factor (or constant dollar FCR), f_{cr} multiplied by the total capital cost. This is a function of the cost of money and the period over which the investment must be paid off (see 2019 NETL report [23]). In this case, it is the lifetime of the plant, plifetime years, plus the construction time, 0 years. Thus, the capital recovery factor is calculated from

$$C(x,N) = \frac{x(1+x)^N}{(1+x)^N - 1} = \left[\sum_{n=1}^{N} \frac{1}{(1+x)^N}]\right]^{-1}, \tag{7}$$

where

$$x = i_c f_c + i_p f_p + (1-t) i_d f_d \tag{8}$$

is the effective cost of money. Here, i_c is the rae of return on common stock, f_c is the fraction of capital form common stock, i_p is the rate of return of preferred stock, f_p is the fraction of capital from preferred stock, i_d is the nterest rate on deb, and f_d is the capital from debt.

From here, the fixed charge rate can be calculated as

$$f_{cr} = \frac{Ck}{(1-t)} - \frac{td}{(1-t)} + t_p + r, \tag{9}$$

where C is the capital recovery factor (7), k is the adjustment for investment tax credit, t is the effective income tax rate, d is the levelized tax depreciation, t_p is the property tax rate and r is the levelized interim replacement cost. For this power plant, see 10 for values. Applying these calculations to this power plant with a lifetime of plifetime yields $f_{cr} = \text{fcr}$.

By multiplying the capital recovery factor, f_{cr} by the total capital cost, C_{99}, the annual capital cost charge is calculated. Annualized Financial Costs are $ 196 M.

N (plant life)	plifetime	t (construction period)	0
i_c	0.1	t_p (property tax rate)	0
i_p	0	x_1 (pre-tax)	0.073
i_d	0.05	x (post-tax)	0.065
f_c	0.45	x_r (real)	0.045
f_p	0	k	1
f_d	0.55	$C(x, N)$	0.077
i (general inflation)	0.02	$C(x_r, N)$	0.061
e (escalation)	0.02	r (interim replacement)	0
e_r (real escalation)	0	f (depreciable fraction of TCC)	0.881
t_s (income tax 1)	0.06	d (levelized depreciation)	0.036
t_f (income tax 2)	0.21	f_{cr}	0.0910
t (effective income tax)	0.257	f_{cr0}	0.0722

Table 10: Values used in calculation of capital recovery factor f_{cr}, the effective cost of money, x, and the fixed charge rate, R.

Cost Category 91 – Escalation

Critical financial concept often used in the planning and analysis of long-term projects, particularly in industries such as energy or construction. This category accounts for the changes in costs over time due to factors like inflation, market dynamics, and changes in labor or material costs. Escalation is the systematic increase in costs over the life of a project. It is distinct from *contingency*, which is used to cover unforeseen costs. Escalation accounts for known, predictable changes in cost such as:

- **Inflation:** General increase in prices and fall in the purchasing value of money.
- **Market Fluctuations:** Changes in costs due to supply and demand dynamics.
- **Labor Cost Changes:** Variations in labor costs due to economic conditions or labor market changes.
- **Material Cost Variations:** Fluctuations in the cost of raw materials.

In many financial estimates, particularly initial cost assessments, Cost Category 91 is often excluded. This exclusion is typically because:

1. **Simplification:** To simplify the early stages of financial planning.
2. **Variability:** Due to the unpredictable nature of some escalation factors.

While often excluded from initial estimates, escalation is crucial in:

- **Business Plans:** For a realistic long-term financial outlook.
- **Financing Proposals:** To present a complete picture to potential investors or lenders.
- **Regulatory-Related Documents:** Where detailed, realistic cost projections are required.

Effective management of Cost Category 91 involves:

1. **Regular Reassessment:** Continuously updating the project's cost estimates to reflect current economic conditions.
2. **Risk Mitigation:** Developing strategies to mitigate the impact of negative cost escalations.
3. **Informed Decision Making:** Using escalation estimates to make informed financial and strategic decisions.

Cost Category 92 – Fees

This Cost Category primarily encompasses the costs associated with annual fees necessary for the operation of such facilities. Fees under Cost Category 92 typically include, but are not limited to, costs related to:

- **Licensed Reactor Processes:** These are fees paid for obtaining and maintaining the licenses required for reactor operation. They cover regulatory compliance and safety standards as set by nuclear regulatory bodies.
- **Nuclear Operating License Fees:** Fees associated with the acquisition and renewal of operating licenses for nuclear facilities. These are critical for legal and safe operation.
- **Regulatory Compliance:** Fees related to ensuring compliance with various environmental, safety, and operational regulations.
- **Inspection and Oversight:** Costs incurred for regular inspections and oversight activities by regulatory authorities to ensure adherence to safety and operational guidelines.

12.3.1 Cost Category 92 – Fees

is an essential financial Cost Category, especially for nuclear facilities, where regulatory compliance and licensing play a critical role. Proper management and allocation of funds for these fees are vital for uninterrupted and legal operation.

Cost Category 93 – Cost of Money

The **cost of money** is akin to the *opportunity cost* of using capital for a specific purpose. When capital is allocated to operating costs, it is an alternative to investing that capital elsewhere. Thus, the cost of money represents the return that this capital could have earned if invested differently.

- Sources of Capital. Capital for operating costs can be sourced either externally or from within the organization (retained earnings). External financing includes loans, bonds, or other forms of borrowing, incurring interest payments. Retained earnings are internal funds saved and not distributed as dividends, carrying an opportunity cost as these funds could be used elsewhere.
- Impact on Operating Costs. In financial Cost Categorying, the cost of money is considered an *indirect cost* of operating. It reflects the financial charges incurred due to the company's capital structure, including both the direct costs (e.g., interest on loans) and the opportunity costs (e.g., using retained earnings).
- Project Financing and Cost of Capital. For large-scale projects like nuclear power plants, the cost of money is crucial in project financing. The strategy (debt vs. equity) and the corresponding costs of capital significantly affect the overall project economics.
- Risk and Return Considerations. The cost of money reflects the project's risk profile. Higher-risk projects lead to higher costs of capital, as lenders and investors demand higher returns for increased risks.
- Long-term Implications. For long-term projects, the cost of money impacts the project's *net present value* (NPV) and *internal rate of return* (IRR). Effective management of the cost of money is essential for the project's long-term financial viability.
- Dynamic Nature. The cost of money changes with market conditions, economic environment, and the organization's financial health. This necessitates continuous monitoring and reevaluation of financing strategies.

In summary, Cost Category 93 - Cost of Money - encompasses the financial implications of the capital used for operating costs. It includes both direct costs (like interest payments) and indirect costs (like opportunity costs of capital). Proper management of this Cost Category is crucial for the economic feasibility and financial health of large-scale projects.

Cost Category 99 – Contingency on Annualized Financial Costs

This Cost Category includes an assessment of additional costs necessary to achieve the desired confidence level for the annualized financial costs not to be exceeded, including schedule uncertainties. Not included here.

12.4 Levelized Cost of Electricity

Net electric power	MW	636.748
Plant availability	%	0.9
Inflation	%	2.5
Lifetime of plant	Years	30
Capital cost, C_{AC}	M$/annum	195.6
Operations and Maintenance Costs, C_{OM}	M$/annum	38.2
Fuel Costs, C_F	M$/annum	1

Table 11: Variables in the LCOE calculation.

Following equation 6, the LCOE is 55.1 $/MWh or 5.5 c/kWh.

A Glossary of terms

ALPHA Accelerating Low-Cost Plasma Heating and Assembly

ARIES Advanced Reactor Innovation and Evaluation Study

ARPA-E Advanced Research Projects Agency - Energy

BOP Balance of plant

CAD Computer-aided design

CBS Cost breakdown structure

DOE U.S. Department of Energy

D-D Deuterium-deuterium

D-T Deuterium-tritium

ETS Energy transfer and storage

FCR Fixed charge rate

FPC Fusion power core

FPP Fusion power plant

FRC Field reversed configuration

HP High pressure

HVAC Heating, ventilation, and air conditioning

I&C Instrumentation and control

IOU Investor Owned Utility

IFE Inertial fusion energy

kWh Kilowatt Hour

LCOE Levelized cost of electricity

LOCA Loss of coolant accident

LP Low pressure

MFE Magnetic fusion energy

MIF Magneto-inertial fusion

MTF Magnetized target fusion

MW Megawatt

MWe Megawatt electric

MWth Megawatt thermal

NEA Nuclear Energy Agency

NIF National Ignition Facility

NRL Naval Research Laboratory

SLC Stabilized liner compressor

SMR Small modular reactor

TAE TAE Technologies

USD U.S. dollars

B LANL NMSBA Report

Dec. 24, 2024

NMSBA December 2024 Initial Report, Plasma Liner Tech-to-Market Effort

Glen Wurden, Simon Woodruff, Feng Chu

Introduction

Since November 2024, a joint effort has been underway between Los Alamos National Laboratory (LANL) and Woodruff Scientific to scope out a next-generation fusion reactor concept based on Plasma Jet Magneto-Inertial Fusion (PJMIF). This collaboration, carried out under the auspices of the New Mexico Small Business Assistance (NMSBA) program and contributing to the ARPA-E Tech-to-Market (T2M) initiative for the Plasma Liner Experiment (PLX), aims to identify key engineering and physics challenges that must be resolved before scaling to a power-producing fusion reactor. While PJMIF is conceptually attractive—particularly because it may provide a potentially simpler, lower-cost path to magneto-inertial fusion—the technology and physics complexities require careful assessment.

Weekly conference calls between LANL, Woodruff Scientific, and occasionally other collaborators (including participants from the Feynman Center and NTTau in England) began informally in November 2024, even though our NMSBA cost code was initiated in early December 2024. During these calls, we have explored parameter ranges, reactor concepts, and the influence of scaling laws on both physics and engineering aspects. We have established a shared Google Drive repository to compile prior research, presentations, and publications on PJMIF and related techniques. This central repository helps ensure that all participants have up-to-date knowledge of the evolving state-of-the-art.

A critical early focus of our work is on system-level considerations. We have leveraged a generic reactor design code developed by Dr. Woodruff (consisting of Python scripts), which allows parametric scans over key input parameters such as fusion yield, driver efficiency, rep-rate, size and geometry of the plasma jets, the number and arrangement of the plasma guns, blanket technology choices, desired electrical output, and more. While we have, for initial bounding studies, assumed robust fusion yield and favorable plasma formation conditions, we recognize that the actual physics of plasma jet formation, compression, and eventual fusion yield must be integrated into future iterations.

This report summarizes our initial findings, focusing on critical physics and technology issues. In particular, we highlight challenges with the plasma guns, blanket technology down selection, the penetrations required by the plasma jets, and the implications of neutronic damage to front-line components.

Background on the PJMIF Concept

The PJMIF concept envisions producing a spherically or cylindrically imploding plasma liner formed from a large array of high-velocity plasma jets. These jets, launched from multiple plasma guns arranged around a chamber, converge and form a continuous liner that symmetrically compresses a magnetized target plasma, potentially reaching fusion conditions. The advantage of such a scheme includes a possibly simpler driver system compared to lasers or pulsed power and the modularity of using multiple identical plasma guns. However, key issues have emerged:

1. **Line-of-Sight and Blanket Penetrations:** For the plasma jets to reach the target, they must have a clear path from the guns to the target plasma center. This implies direct lines-of-sight through any neutron-absorbing blanket. Each open channel reduces the neutron moderation and tritium breeding capability of the blanket system.
2. **Jet Quality and Liner Smoothness:** The fusion performance is highly sensitive to the quality of the imploding plasma liner. Even small non-uniformities can seed turbulence or Rayleigh-Taylor instabilities in the converging plasma, compromising the symmetry needed for effective compression. The discrete nature of multiple plasma guns, each producing a jet of slightly differing density, velocity, and temperature, can lead to undesirable asymmetries.
3. **Component Lifetime in a Harsh Neutron/Gamma Environment:** High-yield fusion events produce copious neutrons and gamma rays. Although the blanket and shielding are designed to mitigate neutron flux, the direct line-of-sight channels needed for plasma jets create potential paths for radiation backscatter. The plasma guns, positioned outside these channels, could face significant neutron-induced damage over time, impacting their operational lifetime and overall reactor economics.
4. **Interaction with Magnetic Fields and Structural Elements:** The presence of guide tubes, final focusing hardware, or background magnetic fields may slow down the plasma jets, alter their density profiles, or cause plasma-material interactions. Such complexities can degrade efficiency, reduce liner quality, or require frequent maintenance.

Literature Review on Plasma-Jet-Driven Magneto-Inertial Fusion (PJMIF)

Introduction

Plasma-Jet-Driven Magneto-Inertial Fusion (PJMIF) is a hybrid approach to fusion that combines elements of inertial confinement fusion (ICF) and magnetic confinement fusion (MCF). It employs an imploding spherical plasma liner, created by merging high-Mach plasma jets, to compress a magnetized target plasma. The goal is to achieve fusion conditions while leveraging standoff designs to minimize hardware destruction and maintain reactor integrity.

Historical Development

The concept of PJMIF emerged in the late 1990s, driven by the challenges in existing fusion methods and the potential of plasma liners as standoff drivers for compressing magnetized plasmas. Early theoretical studies established the parameter space for plasma liners and jets necessary for achieving fusion conditions.

Experimental efforts, such as those on the Plasma Liner Experiment (PLX), began in the early 2010s. These studies demonstrated the feasibility of forming plasma liners using arrays of plasma jets. Advances in plasma-gun technology, including contoured coaxial plasma guns, enabled the controlled generation of jets with the desired speed, density, and uniformity.

Advances in Experimental Studies

- **Plasma Liner Formation:** Experiments focused on merging multiple plasma jets to create uniform spherical liners. Early results indicated significant challenges in achieving uniformity and maintaining high Mach numbers during implosion.
- **Collision Physics:** Studies on jet merging revealed the transition from collisionless to collisional regimes, which is critical for shock formation and energy transfer. These findings validated the theoretical predictions of liner behavior under experimental conditions.
- **Scaling Laws:** Hydrodynamic and radiation-hydrodynamic simulations provided insights into the scaling of liner stagnation pressure and lifetime with initial conditions. These studies established the foundations for optimizing liner parameters.

Current Challenges and Innovations

- **Non-Uniformity Control:** Achieving uniform plasma liners remains a key challenge, as non-uniformities can degrade compression efficiency. Simulation and diagnostic studies aim to address this by refining jet parameters and their synchronization.
- **Liner Energy and Efficiency:** Ensuring that the kinetic energy of the plasma jets translates effectively into compressional heating of the target is crucial. Advances in coaxial plasma gun designs and experimental setups have been instrumental in improving energy coupling.
- **Magnetized Targets:** The integration of magnetized target plasmas, such as spheromaks or field-reversed configurations, with imploding liners is a focus area. Recent experiments explore how the target's magnetic field enhances confinement and suppresses instabilities.

Applications and Future Directions

PJMIF has potential applications in both energy production and high-energy-density physics research. Its scalability and the standoff approach make it a candidate for economically viable fusion reactors. Ongoing work includes:

1. Expanding experimental setups to larger arrays of plasma guns for forming complete spherical liners.
2. Developing subscale experiments to reduce costs while addressing critical physics questions.
3. Exploring advanced diagnostics and computational models for better understanding liner dynamics and energy gain mechanisms.

Conclusion

PJMIF represents a promising path in fusion research, balancing the complexities of ICF and MCF while introducing innovative plasma liner-driven methods. Despite its early stage, the ongoing experimental and theoretical efforts underline its potential to overcome the limitations of traditional approaches and contribute significantly to the realization of practical fusion energy.

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Critical Physics and Technology Issues

1. Plasma Gun Design and Efficiency

The efficiency of plasma guns—both in terms of converting electrical energy into kinetic plasma energy and in achieving a high-quality, stable jet—is paramount. Current considerations indicate that achieving sufficiently uniform and repeatable plasma jets at repetition rates consistent with commercial power generation (e.g., multiple shots per second) is non-trivial. Several issues arise:

- **Gun Electrode Erosion:** High-speed plasma jets inevitably cause electrode erosion, especially if the system operates at high repetition rates. Electrode wear leads to metallic impurities in the plasma, reducing liner integrity and possibly contaminating the fusion plasma. It also limits gun lifetime and increases maintenance costs.
- **Neutron-Induced Embrittlement:** Even if the guns are placed outside a thick neutron shield, line-of-sight channels allow some fraction of the neutron flux to impinge on the front-end structures. Over time, materials suffer embrittlement, swelling, and loss of mechanical integrity. This factor directly impacts gun design since it dictates how frequently guns must be replaced, how thick the shielding can be without overly compromising jet penetration, and what materials can be used for gun electrodes and support structures.
- **Driver Efficiency and Energy Management:** The energy required to accelerate the jets to the necessary velocity (tens of km/s) must be minimized. Improved magneto-plasma-dynamic (MPD) gun designs, better coaxial plasma gun configurations, or novel solid-state switching systems could improve efficiency. However, these solutions must not only address energy efficiency but also do so within the constraints of a neutron-rich, radiation-damaged environment.

2. Blanket Technology Downselection and Jet Penetration

A key engineering challenge arises from the blanket design. The blanket must perform three critical functions: (1) moderate and absorb neutrons to produce tritium via breeding reactions in lithium-containing materials; (2) shield external structures and personnel from radiation; and (3) efficiently extract heat for conversion into electricity. Penetrations for plasma jets complicate this:

- **Tritium Breeding Ratio (TBR) Maintenance:** Every line-of-sight channel for a plasma jet reduces the effective breeding volume. If too many channels are needed, it becomes difficult to maintain a TBR greater than 1.0, which is essential for a self-sustaining fusion cycle.
- **Neutron Streaming and Localized Damage:** The penetrations create pathways for neutron streaming. This can lead to localized hot spots in the reactor structure behind the blanket,

especially near the gun ports. Over time, this may reduce the reliability of critical components, including diagnostics, feed systems, and structural supports.

- **Guide Tubes and Plasma-Material Interactions:** One proposed solution is to introduce guide tubes that run through the blanket, allowing plasma jets to travel in a dedicated channel. However, these guide tubes pose their own challenges. They might cause frictional slowing of the plasma jets, introduce impurities from ablated wall materials, or become coated with debris from previous fusion shots. Over many shots, the tube surface could roughen, adding turbulence and reducing the effective velocity and coherence of the jets. If the guide tubes are conductive, they might also interact with any applied or residual magnetic fields, adding complex magnetohydrodynamic (MHD) effects that need to be understood and mitigated.

3. Reactor Geometry: Spherical vs. Cylindrical Implosions

An open design question is whether spherical implosions are strictly required for PJMIF, or if cylindrical or even pseudo-spherical (polygonal) arrays of jets might suffice. Spherical implosions are believed to yield the most symmetrical compressions, critical for achieving the high densities and temperatures needed for net fusion gain. However:

- **Spherical Designs and Complexity:** Spherical arrays require large numbers of jets arranged in three-dimensional patterns, complicating both the blanket design and the maintenance of uniform jet parameters. More jets mean more penetrations, lower TBR, and a greater fraction of the reactor wall area dedicated to gun ports rather than neutron moderation.
- **Cylindrical or Linear Geometries:** These may reduce the complexity of jet alignment and the number of guns needed, but could compromise compression symmetry. While cylindrical implosions might still achieve high compression ratios, they might not reach the levels required for a positive net energy gain without more complex shaping or more intense magnetic fields.
- **Magnetic Field Configuration and MHD Stability:** The presence of background magnetic fields, used to trap α-particles or stabilize the target plasma, adds complexity. Plasma jets must penetrate these fields without significant energy loss or deflection. Different geometrical configurations of the implosion may alter how easily the jets navigate or disrupt the field configuration.

Plasma Guns

One of the critical engineering challenges in plasma jet fusion concepts, such as those pursued on the Plasma Liner Experiment (PLX), is the design, placement, and longevity of the plasma guns that produce and launch the high-velocity jets. These guns are typically complex, multi-component devices that incorporate insulators, electrodes, and energy storage elements. Their cost is influenced not only by the materials required—such as copper nozzles and other specialized alloys—but also by how well the guns are protected from the harsh fusion environment, including neutron and gamma radiation damage over extended operational periods.

A key design consideration for the guns is how and where to store the energy required for each plasma shot. The energy is often delivered from capacitor banks or Pulse Forming Networks (PFNs) that must be placed close to the gun to minimize inductance. Keeping the energy storage close reduces the inductive losses and the complexity of long high-voltage cable runs. On the other hand, placing large, sensitive capacitor banks near the reaction chamber increases their exposure to radiation. This can lead to compromised component lifetimes and may necessitate thicker shielding, more robust materials, or more frequent maintenance.

The placement of insulators and the overall geometry of the gun assembly are strongly influenced by the need for shielding and easy replacement. Insulators, often critical weak points, must be shielded from the neutron flux generated during fusion events. Estimates suggest that at least a meter of shielding may be required to protect sensitive gun components, including insulators, from direct neutron exposure. If guns are mounted further away—say at a distance of five meters—this reduces the neutron fluence on the gun, but at the cost of adding more high-voltage cabling and therefore more complexity, potential inductive losses, and points of failure.

Longevity and maintenance intervals are major economic considerations. A 1 Hz firing rate translates to 3,600 shots per hour, or over 32 million pulses per year, if operated continuously. Achieving a gun lifetime of this magnitude is challenging. Current designs, constructed with materials like brass or copper nozzles, may not be able to withstand the cumulative neutron fluence, erosion from repeated plasma shots, and mechanical stresses associated with repetitive pulsed operation without substantial performance degradation. While short-term experiments and Doug Witherspoon's publications provide valuable data on gun performance and materials endurance, scaling from laboratory testing (hundreds or thousands of shots) to a power plant scenario (tens of millions of shots per year) presents a daunting materials and engineering challenge. Neutrons escaping through these channels can:

- **Degrade Gun Materials:** Structural materials may suffer from neutron-induced swelling, embrittlement, and transmutation over time. This shortens the mean-time-to-failure of gun components and drives up operational costs.
- **Impact on Lifetime and Maintenance Cycle:** The cost model for the reactor depends heavily on how often one must replace or refurbish guns. If the neutron flux along these channels significantly reduces gun lifetime, the reactor's economics and capacity factor will be severely impacted. Reliability and operational availability become key figures of merit.
- **Radiation-Hardened Designs:** To mitigate these issues, more advanced materials (e.g., tungsten alloys, or specialized steels) or sacrificial liners in the gun ports may be required. These would be replaced periodically, adding complexity to the reactor maintenance schedule.

Gun efficiency is another non-negligible factor. Current designs indicate that around 20% of the stored electrical energy is effectively converted into kinetic energy of the plasma jet. Improving this efficiency could reduce the required driver energy, ultimately lowering operational costs and relaxing some engineering constraints.

In summary, the plasma guns represent a critical nexus of physics and engineering challenges. They must be engineered to reliably deliver plasma jets at high repetition rates, be positioned and shielded against harsh neutron and gamma radiation, incorporate accessible energy storage without incurring large inductive penalties, and demonstrate long operational lifetimes under power-plant-level conditions. Balancing these factors—cost, complexity, placement, shielding, lifetime, and efficiency—will be central to the success of plasma jet-driven magneto-inertial fusion approaches.

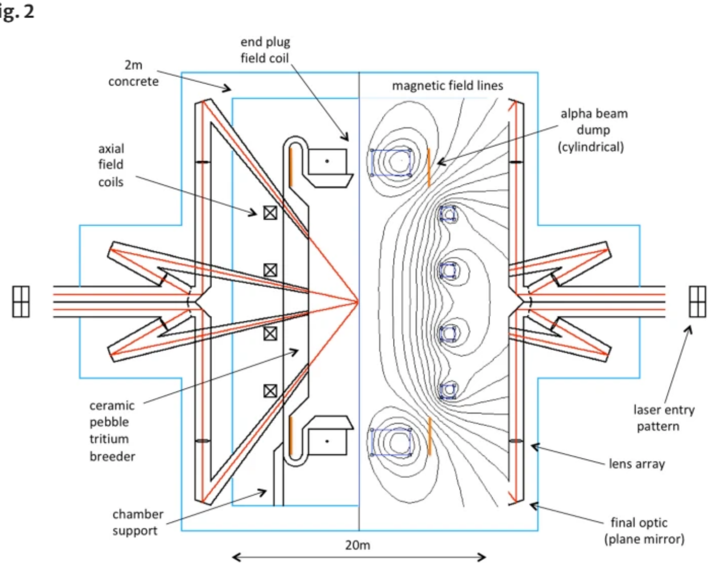
Incorporation of Colliding FRC Targets and Magnetic Guide Fields

As our scoping studies progress, one particular target plasma concept under consideration is the use of colliding Field-Reversed Configurations (FRCs) as the central "seed" targets for compression by the plasma liner. FRCs, being compact toroidal plasmas with closed magnetic field lines and no material center conductor, offer an attractive target for a magneto-inertial fusion scenario: they inherently possess a self-organized magnetic field structure that can help confine charged fusion products, and they can be formed and translated relatively easily within linear or cylindrical geometries.

FRC Formation and Collision:
In a PJMIF reactor scenario, two or more FRCs can be created by plasma injectors or coaxial FRC launchers located at opposite ends of a cylindrical chamber. Once formed, these FRCs are translated and accelerated towards each other, eventually colliding and merging in the center of the reaction chamber. The collision process can produce a more robust, high-density target plasma with improved magnetic confinement characteristics. The merged FRC would then serve as the target to be further compressed and heated by the imploding plasma liner formed from the array of jets.

Initial modeling suggests that colliding FRCs might produce a more stable magnetic configuration at the center of the reactor chamber, potentially improving overall fusion performance compared to a simple magnetized plasma blob. The ability of FRCs to maintain strong poloidal fields without an internal coil opens possibilities for integrating them seamlessly into a PJMIF reactor design.

Fig. 2



Overview (Elevation) of direct drive target chamber

Figure 1. Example of a magnetic field imposed on a target chamber.

Magnetic Guide Fields in a Cylindrical Chamber:
To support the formation, translation, and collision of FRCs, we are exploring the addition of a controlled background magnetic field (a guide field) within a cylindrical reaction chamber. This guide field would be established by external magnets (e.g., superconducting coils) arranged around the chamber, providing a magnetic "track" that helps maintain the integrity and alignment of the FRCs as they move towards their collision point.

Key considerations include:

1. **Field Strength and Geometry:**
   The guide field must be strong enough to control FRC motion without excessively constraining the plasma jets that form the liner. A weak to moderate axial field may assist in FRC stability and reduce MHD instabilities during translation. However, too strong a field could impede the inward motion of the plasma liner jets or require them to overcome significant magnetic pressure, thus reducing efficiency.
2. **Interplay with Plasma Jets and Liner Formation:**
   The presence of a structured background field could influence how plasma jets propagate into the chamber. Jets entering along paths that intersect these fields may experience MHD effects, altering jet trajectory, velocity, or uniformity. We must ensure that the magnetic field configuration does not inadvertently slow or distort the jets, which would degrade liner symmetry and compression efficiency.
3. **Material and Mechanical Considerations:**
   Integrating a set of superconducting or high-field magnets around the chamber introduces engineering complexity. These magnets must be shielded from neutrons and high-energy gamma rays and remain operational under intense radiation conditions. Cooling, structural support, and maintenance access around these magnets all become integral parts of the reactor's mechanical design.
4. **Synergy with Blanket and TBR Maintenance:**
   Just as with plasma jet ports, magnet placement may influence blanket geometry. While magnets would likely be placed outside the primary neutron-absorbing region, they still require accessible maintenance paths and may affect how the blanket can be arranged. Ensuring that adding magnets does not further reduce the effective blanket coverage or TBR must be a priority.

Advantages and Challenges of FRC Targets:
Using FRCs could provide a more favorable starting point for the compression phase. An FRC-based target might reduce the required liner symmetry, tolerating certain non-uniformities due to its own magnetic confinement. It may also enable lower implosion velocities or pressures, potentially relaxing some engineering constraints on the plasma guns.

However, the formation and collision of FRCs at reactor scale remains an active area of research. The reproducibility of FRC properties (density, temperature, magnetic flux) and the consistency of their collision and merging process must be demonstrated. Additionally, translating this concept from a laboratory experiment to a power reactor scenario demands exploring how the FRC's lifetime, stability, and performance scale with size, magnetic field configuration, and repetition rate.

Path Forward for FRC Integration:
Our near-term plan includes conducting simulations and smaller-scale experiments (e.g., at PLX or related facilities) to evaluate how a background guide field and colliding FRCs interact with plasma jet compression. Leveraging computational tools (e.g., MHD and kinetic simulations) will help map out parameter space for the axial guide field strength, FRC density and geometry, and plasma jet parameters.

As we refine our baseline reactor model, we will incorporate FRC targets as an option in Dr. Woodruff's reactor design code. By varying input parameters related to the FRC (such as formation energy, merging velocity, and magnetic field configuration), we will assess its impact on reactor performance, cost, maintenance schedules, and overall feasibility. Ultimately, if colliding FRCs and guide fields prove advantageous, they could become central elements of the PJMIF reactor design, potentially enhancing plasma stability and fusion gain while mitigating some of the physics and engineering challenges highlighted in previous sections.

Discussion

Achieving high fusion gain in a magneto-inertial fusion (MIF) concept is highly dependent on the geometry and symmetry of the compression process. A spherical implosion is often preferred for uniform compression and to minimize instability seeds, and it naturally fits with a spherical blanket configuration designed for effective neutron moderation and tritium breeding. However, when considering Field-Reversed Configurations (FRCs) as the target plasma, the use of a spherical geometry becomes problematic. FRCs, with their toroidal topology and closed field lines, lend themselves more readily to cylindrical compression. In such a cylindrical setup, compression occurs radially and along the axis ("2.5D" compression), relying on the tension in the magnetic field lines to constrain the plasma and prevent it from escaping axially. This approach typically involves launching two FRCs from opposite ends of a cylindrical chamber and allowing them to collide and merge at the center, forming a more robust target. To accomplish this reliably, a guide magnetic field is needed. Without a pre-existing axial magnetic field, controlling and compressing the FRC is significantly more challenging.

Slowing down and stabilizing the liner compression to match the slower dynamics of forming and merging FRCs could help integrate them as targets, but this may reduce the peak pressures achieved and potentially the net energy gain. This stands in contrast to approaches pursued by companies like Helion or TAE, which drive FRCs with neutral beams and other steady-state methods. While using plasma liners to compress such targets is conceptually appealing, the added complexity of generating stable, repeatable FRCs has prompted some developers (e.g., General Fusion) to switch to other target types, such as spherical tokamak-like configurations or other compact toroids that are simpler to produce.

Regardless of target choice, shaping the imploding liner can help direct compression forces to prevent plasma escape. In a cylindrical setup, thinning the liner at the ends can accelerate compression at those locations, effectively “plugging” the ends. This prevents plasma from squirting out along the axis, much like squeezing a tube of toothpaste at both ends to force it to compress in the middle.

Engineering factors strongly influence these physics decisions. The timescale of the implosion, the need to clear the chamber between shots, and the achievable repetition rate all constrain what can be done practically. A repetition rate of once every ten seconds is considered relatively manageable, especially with methods like flowing liquid walls to quickly clear debris. Pushing to higher rates—such as ten shots per second—places stringent demands on pumping speed, chamber design, and driver technology, especially for large chambers on the order of ten meters in diameter. Achieving compression ratios of around 15:1 is currently seen as more realistic than more extreme compressions, given materials limits and engineering complexity.

In the near term, discussions and collaborations with experts like Petros Tzeferacos—who leads computational modeling efforts at the FLASH Center—may provide further insights. His visit, anticipated in January, could help refine models and identify practical paths to implement these physics concepts at scale. Ultimately, the interplay between plasma physics, geometry, materials, and engineering constraints shapes the decision-making process, guiding the selection of target concepts, liner configurations, and operational strategies for next-generation MIF systems.

Path Forward and Ongoing Studies

In our weekly calls and ongoing research activities, we have identified several immediate tasks:

1. **Parametric Sensitivity Studies:** Using Dr. Woodruff’s reactor design code, we will systematically vary parameters such as number of jets, jet velocity, and blanket thickness. This will help us understand how sensitive the reactor economics and net electrical output are to these variables.
2. **Neutronics and Tritium Breeding Analysis:** A detailed neutronics simulation (using tools like MCNP or similar codes) is needed to model the streaming paths through jet penetrations. We must quantify the trade-offs between adding more jets (for smoother liners) and reducing blanket coverage (lower TBR).
3. **Material Testing and Irradiation Studies:** Early-stage materials testing, including irradiation experiments in accelerator-based neutron sources, can help us understand which materials might offer better longevity for guns and guide tubes.
4. **Plasma-Jet Interaction with Magnetic Fields:** Experiments at PLX and other test facilities will be used to study plasma flow through guide tubes and across magnetic field lines. Understanding MHD effects is crucial before scaling up to a reactor environment.
5. **FRC target considerations:** Develop at the conceptual level a FRC target formation scheme that can provide the target plasma in the center of the device as the jets are imploding.

6. **Engineering Mockups and Costing Models:** Finally, we plan to develop more detailed costing models that incorporate maintenance schedules, replacement costs for guns, and variable blanket configurations. The goal is to produce a baseline reactor model that can guide future ARPA-E T2M efforts and inform decision-making on downselecting concepts for more intensive research and development.

Conclusion

The PJMIF concept, while promising, faces non-trivial challenges that must be addressed for it to mature into a viable reactor system. Critical among these are maintaining plasma jet quality and efficiency, ensuring sufficient tritium breeding and blanket effectiveness despite multiple jet penetrations, and mitigating neutron damage to critical components like plasma guns.

Our initial efforts, supported by the NMSBA program and integrated into the ARPA-E T2M initiative, have begun to reveal the complexity of the PJMIF reactor design space. As we move forward, we will refine our understanding of the interplay between physics parameters (fusion yield, jet quality, geometry) and engineering constraints (blanket coverage, neutron shielding, materials resilience). Ultimately, the success of PJMIF as a fusion reactor concept depends on how effectively these challenges can be tackled through informed design choices, advanced materials, and innovative operational strategies.

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